

Statistical models as natural transformations: a bridge from the McCullagh–Brøns hexagon to Markov categories

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August 2026

Abstract

McCullagh’s 2002 categorical definition of a statistical model, sharpened by Brøns into a hexagonal diagram of categories and functors, identified naturality as the mathematical content of the informal requirement that the meaning of a parameter be “retained” across experimental designs. The programme supplied a language but not a calculus: no ambient categorical probability theory existed in which the naturality condition could be computed with, composed, or mechanically verified. We show that the Markov-categorical framework of Fritz, Cho–Jacobs and Perrone supplies exactly this calculus. Our main result recasts a McCullagh–Brøns model as a natural family of morphisms in the Kleisli category of the Giry monad, deterministic on the parameter side; under this translation McCullagh’s criterion for a meaningful subparameter becomes a naturality condition, hence a type-level property under the functorial encoding, and Tjur’s weaker elementary criterion, invariance under design reduction, is recovered as naturality restricted to the insertion morphisms. We prove the statement in full for finite (discrete) models, state the measurable case with its precise additional hypotheses, and formulate the coherence problem for finite designs in the cohomology of a diagram of algebras in the sense of Gerstenhaber–Schack and Baues–Wirsching, connecting it to recent work of Caputi and the author. We show that the obstruction so obtained is necessarily *class-relative*, the unrestricted class vanishes for every quantity, by construction, and discharge the resulting statement by an exact computation for the one-way layout, which certifies the non-meaningfulness of the marginal dispersion relative to the class of information-bearing recalibrations for scale-confounded designs, recovers the within-group dispersion as a correction by pure linear algebra, and exhibits H^1 of the design scheme as a computable invariant of independent interest.

MSC 2020: 62A01 (primary); 18M05, 60A05, 18G90 (secondary).

Keywords: statistical model; Markov category; Giry monad; natural transformation; meaningfulness; design category; Baues–Wirsching cohomology.

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1 Introduction

A recurring observation is that a small canon of unifying principles holds statistics together: likelihood, sufficiency, conditioning, ancillarity, invariance, exchangeability. Each principle is made precise inside its own apparatus (measure-theoretic probability, decision theory, group actions), and their mutual coherence is maintained by the trained intuition of the field rather than by a single calculus in which they are theorems. The best-known sustained attempt at such a calculus is Birnbaum’s [6], which formulates sufficiency and conditionality as axioms on evidential meaning and argues that together they force the likelihood principle. That its status remains disputed sixty years on, over what the conditionality principle *says*, rather than over any computation [29, 12], is itself evidence that the principles have lacked an ambient formalism in which such a question is settled. McCullagh’s “What is a statistical model?” [30] diagnosed one root of this state of affairs: the standard definition of a statistical model, a parametrised family of probability measures on a fixed sample space, is silent about how the family must transform when the experimental design changes, and therefore cannot even *state*, let alone enforce, the requirement that a parameter retain its meaning across designs. His remedy was categorical: designs form a category, sample spaces and parameter spaces become functors on it, and a model is a natural transformation. Brøns, in his discussion of the paper [9, pp. 1280–1282], organised these data into a hexagonal diagram of categories and connecting functors, and observed that the resulting notion of model is equivalent, by a general comma-category argument, to a diagram in a fixed “category of statistical models”. In the same passage he records that this category is too small for the purpose, to include all statistical transformations, he writes, “Markov kernels must be allowed”, and then sets the point aside [9, p. 1282]. The present paper takes it up. Tjur, in the same discussion volume [36], isolated the operative consequence, and did so *without* categorical language, which he judged convenient rather than necessary: fixing an infinite population of covariate–response pairs, he called a rule assigning a parameter, a function of the unknown distribution, to each finite sample of covariate values, repetitions allowed, a *parameter function*, and such a function *meaningful* when it is invariant under the formation of marginal models, when the design is reduced by removing some of the sampled x ’s, the parameter of the marginal distribution must agree with the parameter for the bigger sample. The removals are of sample points, not of points of a covariate space: before the criterion is imposed, his parameter functions include the sample size n itself. His non-example is the design-averaged expectation $\alpha + \beta \bar{x}$, with $\bar{x}$ the average over the sample one happened to draw; he attributes the analogous failure for the correlation coefficient to McCullagh. The categorical form of this criterion, naturality, is McCullagh’s own [30, §4.5], his rejoinder reads Tjur’s remarks as explaining what is meant by a natural parameter in regression [30, p. 1304]: a *natural subparameter* is a natural transformation of functors on the design category, and inference is meaningful only for natural parameters. The two formulations are stated over different morphism classes, Tjur’s over design reduction alone, McCullagh’s over the full design category, and neither source asserts that they coincide as mathematical requirements.

The programme is widely admired and rarely used. We contend that the reason is structural rather than sociological: in 2002 the codomain of the construction was bare **Set**, so naturality could be written down, but nothing could be *done* with it, there was no ambient categorical probability theory in which conditioning, sufficiency, or almost-sure reasoning were native operations and no technology for mechanically verifying naturality. Both missing ingredients have since appeared, independently and for unrelated reasons. Markov categories [14, 11, 32, 15] turn Markov kernels into the morphisms of a symmetric monoidal

category with enough structure to state and prove, synthetically, the Fisher–Neyman factorisation, versions of sufficiency and ancillarity, and de Finetti-type theorems.

This paper does the obvious-in-retrospect thing: it moves the codomain of the McCullagh–Brøns construction from **Set** to the Markov category **Stoch** (finite case: **FinStoch**) and records what becomes true, what becomes provable, and what becomes computable. The contributions are deliberately modest in scope and precise in status:

1. Section 2 gives a self-contained account of the McCullagh–Brøns framework, including a complete statement and proof of the comma-category equivalence (Proposition 2.7), which in the original discussion is asserted with a sketch.
2. Section 3 recalls the Markov-categorical background actually used.
3. Section 4 states and proves the bridge in the finite case (Theorem 4.1), states the measurable case with its exact additional hypotheses (Theorem 4.3), and is explicit about the fact that those hypotheses are a genuine restriction relative to McCullagh’s **Set**-valued formulation (Remark 4.4). It also records, opening §4.3, that the bridge is, in disguise, a representability-plus-copower computation (Remark 4.7), and uses this to reformulate what we call the *arrow-category identification*: the open question whether Brøns’ category of statistical models is equivalent to a category built canonically from **Stoch**. The answer is that, on the objects whose parameter set and sample space satisfy two explicit regularity hypotheses (automatic in the finite case), it is equivalent to the arrows of **Stoch** *with copower domain* (Proposition 4.11); for the measurable comma category the same holds without the copower restriction (Remark 4.13); the passage between the two is an adjunction exhibiting Brøns’ category as a coreflective subcategory of its measurable analogue, so that the copower restriction is the non-invertibility of a counit and nothing more (Proposition 4.14 and Corollary 4.15).
4. Section 5 defines Tjur meaningfulness precisely, proves its characterisation as a naturality condition under the bridge (Proposition 5.4), and reduces its verification over finite designs to finitely many matrix equalities (Corollary 5.7).
5. Section 6 formulates the coherence problem for finite designs cohomologically: a candidate design-indexed quantity is correctable, within a prescribed admissible class, to a natural (hence meaningful) one iff an explicit obstruction cochain in the cohomology of a diagram of algebras à la Gerstenhaber–Schack [18]/Baues–Wirsching [4] is the coboundary of a correction in that class. The equivalence is proved (Proposition 6.4), together with the observation that forces the obstruction to be read relative to an admissible correction class. The one-way layout is then carried out in full (Examples 6.6–6.8).
6. Section 7 delimits the scope. Several statements adjacent to the results below (paradigm unification, nonparametric limits) are formulable in this framework but are not proved here; the boundary is drawn explicitly rather than left blurred.

Related work. The closest prior work is Patterson’s thesis [31], which develops an algebraic theory of statistical models with machine representation in view; the present paper differs in taking the design-indexed (McCullagh–Brøns) structure, rather than the model-theoretic signature, as primitive and specifically targeting the naturality/meaningfulness axis. Fritz [14] proves synthetic sufficiency theorems for a single fixed sample space, *inside* the hom-sets of one Markov category [14, Def. 14.1]; McCullagh–Brøns naturality is a condition *across* a diagram of such hom-sets; the two are complementary, and their combination (fibred over the design category) is exactly the structure studied here. Jacobs’ channel-based probability [24], the synthetic de Finetti theorem [15], the representable-category comparison of statistical experiments [16] and the dilational axiomatics of [17] supply techniques we expect to import in later work. Categories of Markov kernels have also been put to work well outside inference proper: Sergeant-Perthuis develops rigorous statistical mechanics compositionally, with statistical systems presented as representations of posets and their Gibbs phases as invariants of those representations, in a setting that couples

measurable maps with Markov kernels [34, 35]. That the same category of kernels underlies both that setting and the design-indexed one studied here is evidence of its generality: the two settings are unrelated here, and no correspondence is claimed. Helland’s work on statistical symmetry [21] pursues the invariance principle by group-theoretic rather than categorical means; the compositional approach to learning of [13] is adjacent to, but distinct from, the design-indexed structure studied here.

Two further observations in Brøns’ discussion bear on the shape of what follows, and we record them here because they are easily mistaken for departures from the 2002 framework when they are in fact requests made within it. First, having recognised a model as a diagram in his category of statistical models over the basic category, he remarks that no evident reason confines the theory to base categories of that particular form, and proposes admitting diagrams over arbitrary categories, which would in particular subsume the passage to subcategories [9, p. 1282]. Extending the indexing beyond the comma-category shape of Definition 2.1 is therefore continuation rather than importation. Second, he observes that the product and repetition constructions of [30] are best treated with monoidal categories, monoidal functors and monoidal natural transformations, noting that $\mathcal{P}$ carries a monoidal structure given by the product of measures [9, pp. 1282–1283]. The codomain adopted here is monoidal in exactly that sense; the copy–discard structure recalled in §3 is a strengthening of it, and is what makes conditioning and independence expressible rather than merely the product of models.

2 The McCullagh–Brøns framework

We fix the notation of Brøns’ discussion [9], in the form used throughout the sequel. All categories in this section are ordinary (locally small) categories.

Definition 2.1 (Design data). The framework is parametrised by:

1. a category $\mathbf{U}$ of *statistical units* (typical objects: finite sets of experimental units; typical morphisms: injections, or restriction to subsamples);
2. a category $\mathbf{\Omega}$ of *covariate spaces*;
3. a category $\mathbf{V}$ of *response scales* (typical morphisms: measurable surjections implementing coarsening of the response, affine rescalings, etc.);
4. the category $\mathbf{S} := \mathbf{V} \times \mathbf{U}^{\text{op}}$ of *sample space indices*, together with a functor $\Gamma: \mathbf{S} \rightarrow \mathbf{Meas}$ sending (v, u) to the measurable space of v -valued responses on the units u (canonically $\Gamma(v, u) = v^u$ with product structure when v is a measurable space and u a finite set);
5. the category of *designs*

$$\mathbf{D} := (J_{\mathbf{U}} \downarrow J_{\mathbf{\Omega}}),$$

the comma category of a pair of functors $J_{\mathbf{U}}: \mathbf{U} \rightarrow \mathbf{C}$, $J_{\mathbf{\Omega}}: \mathbf{\Omega} \rightarrow \mathbf{C}$ into a common ambient category (in the basic examples $\mathbf{C} = \mathbf{Set}$ and both J ’s are the underlying-set functors). An object of $\mathbf{D}$ is thus a triple (u, ω, x) with $x: J_{\mathbf{U}}u \rightarrow J_{\mathbf{\Omega}}\omega$ a *design map* assigning covariate values to units; a morphism $(u, \omega, x) \rightarrow (u', \omega', x')$ is a pair of morphisms $\varphi: u \rightarrow u'$ in $\mathbf{U}$, $\psi: \omega \rightarrow \omega'$ in $\mathbf{\Omega}$ with $x' \circ J\varphi = J\psi \circ x$. We write $\text{pr}_{\mathbf{U}}: \mathbf{D} \rightarrow \mathbf{U}$ and $\text{pr}_{\mathbf{\Omega}}: \mathbf{D} \rightarrow \mathbf{\Omega}$ for the projections;

6. a concrete category $\mathbf{K}$ of *parameter objects*, with faithful functor $U_{\mathbf{K}}: \mathbf{K} \rightarrow \mathbf{Set}$, and a contravariant *parameter functor* $\Theta: \mathbf{\Omega}^{\text{op}} \rightarrow \mathbf{K}$.

Remark 2.2. Contravariance of Θ encodes the statistical fact that enlarging the covariate space enlarges (or reparametrises) the parameter, while a morphism of covariate spaces induces a map of parameters in the opposite direction; the regression examples of [30, §6] are of this shape. Contravariance of the $\mathbf{U}$ factor in $\mathbf{S}$ encodes that restricting to a subsample induces a (projection) map of sample spaces in the opposite direction.

Let $\mathcal{P}: \text{Meas} \rightarrow \text{Set}$ denote the functor sending a measurable space to its set of probability measures and a measurable map to the pushforward. (This is the underlying-set composite of the Giry monad $\mathcal{G}: \text{Meas} \rightarrow \text{Meas}$ [20].) Note that $\mathcal{P}$ is **Set**-valued: a map $\Theta_0 \rightarrow \mathcal{P}(S)$ is a family of probability measures on S indexed by Θ_0 , with no measurability imposed in the index. This is the gap the bridge of §4 has to close, vanishing when the parameter object is discrete, and otherwise an added hypothesis (Theorem 4.3).

Definition 2.3 (McCullagh–Brøns model). With the data of Definition 2.1, form the two composite functors $V \times D^{\text{op}} \rightarrow \text{Set}$:

$$L := U_K \circ \Theta \circ \text{pr}_\Omega^{\text{op}} \circ \text{pr}_2 \quad \text{and} \quad R := \mathcal{P} \circ \Gamma \circ (\text{id}_V \times \text{pr}_U^{\text{op}}),$$

where $\text{pr}_2: V \times D^{\text{op}} \rightarrow D^{\text{op}}$ is the projection. A *statistical model* is a natural transformation

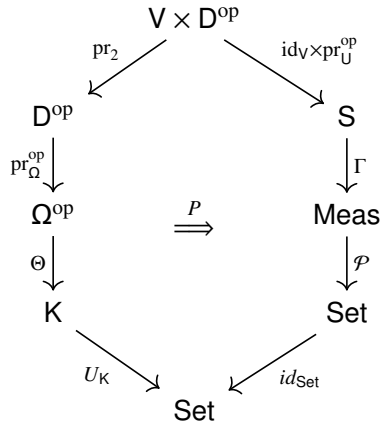
$$P: L \Rightarrow R.$$

Concretely: for every response scale v and every design $d = (u, \omega, x)$, a map

$$P_{v,d}: U_K \Theta(\omega) \longrightarrow \mathcal{P}(\Gamma(v, u)), \quad \theta \longmapsto P_{v,d}(\cdot \mid \theta),$$

assigning to each parameter value a probability measure on the sample space, such that for every morphism $(\alpha, (\varphi, \psi)^{\text{op}})$ of $V \times D^{\text{op}}$ the evident square commutes.

The six categories U, V, Ω, S, D, K and the functors between them, with **Meas** and **Set** as ambient codomains, assemble into Brøns’ hexagonal diagram; a model is a natural transformation filling the hexagon between its two composite paths:



Remark 2.4 (Brøns’ packaging and McCullagh’s own definition). The hexagon is Brøns’. McCullagh’s definition in [30, §4.2] holds the response scale *fixed* (§4.7 of the same paper lets it vary): with the scale fixed, the parameter is a contravariant functor $\Omega \rightarrow K$, and a model is a natural transformation of functors on D alone. The product categories $S = V \times U^{\text{op}}$ and $V \times D^{\text{op}}$ are McCullagh’s [30, §4.7, p. 1240]; Brøns restates them in his own notation [9, p. 1280] and then packages Θ as a functor on Ω^{op} alone. Three differences are worth recording, since we use Brøns’ formulation throughout.

- (i) *The V slot is additional data.* In the hexagon V varies while Θ is defined on Ω^{op} alone, so a non-identity response-scale morphism constrains the model at a fixed parameter value. In McCullagh’s own worked construction the parameter of a linear model is a subrepresentation $\Theta_\omega \subseteq v^\omega$ of the standard representation [30, p. 1235], and therefore *does* depend on the response scale. The two agree exactly when v is held fixed, which is the hypothesis of McCullagh’s §§4.2–4.6; the discrepancy is otherwise real, and is what Remark 2.12 measures.

- (ii) *Morphism restrictions.* McCullagh takes the morphisms of $\mathbf{U}$ and $\mathbf{\Omega}$ to be injective [30, §4.1]; Brøns judges such restrictions dispensable and does not impose them [9, p. 1281]. Definition 2.1 follows Brøns for $\mathbf{\Omega}$; for $\mathbf{U}$, every example below takes $\mathbf{FinSet}_{\text{inj}}$, and the verifications of §2.2 do use the injectivity of unit maps (see the remark following (1)), so that restriction is not dispensable there. The choice for $\mathbf{\Omega}$ is not cosmetic: it fixes the class over which naturality is quantified, and hence the gap between McCullagh’s meaningfulness criterion and Tjur’s, as §5 makes precise.
- (iii) *Nothing below depends on the difference.* Every statement in the sequel is a statement about the naturality squares of Definition 2.3, and for fixed v these are exactly McCullagh’s squares of [30, p. 1236]. Where the V slot does work of its own, we say so.

Example 2.5 (One-way layout). This is the running example of the paper. We give it in the same detail as the instantiations of §2.2, which it precedes only because it is the simplest of the three; the verification of naturality is given at the end of §2.2, once the naturality squares have been displayed.

The experiment, in words. A single *factor* is applied to each experimental unit: every unit receives exactly one of finitely many mutually exclusive interventions, its *treatment*. There is no blocking, no repeated measurement, and no explanatory variable other than the treatment received, this is what *one-way* means. A *treatment label set* is a finite non-empty set ω whose elements name the available treatments (equivalently, the *levels* of the factor). Only the naming is retained: ω carries no order, no metric and no algebraic structure. That is precisely why the covariate category below consists of bare finite sets rather than of vector spaces as in Example 2.8 below; the passage between the two descriptions is dummy coding, and is Remark 2.18.

1. **Units.** $\mathbf{U} = \mathbf{FinSet}_{\text{inj}}$: an object u is an index set for the experimental units actually enrolled; a morphism $\varphi: u \hookrightarrow u'$ is the inclusion of a subsample. We write $n = |u|$.
2. **Covariate spaces = treatment label sets.** $\mathbf{\Omega} = \mathbf{\Omega}_{\text{lab}}$, the full subcategory of $\mathbf{FinSet}$ on the finite non-empty sets: objects are treatment label sets, morphisms are *all* maps $\psi: \omega \rightarrow \omega'$ of such sets. Four kinds of morphisms carry the statistical readings, and every ψ is a surjection followed by an injection, so the first two generate:
 - ψ *surjective*: *merging* of treatments. The fibres $\psi^{-1}(b)$, $b \in \omega'$, are the blocks of treatments declared indistinguishable.
 - ψ *injective and not bijective*: *selection* of a proper subfamily of treatments, i.e. the restriction of the experiment to some of its groups. When ψ is a proper inclusion $\omega' \subsetneq \omega$, the surviving labels carried unchanged, we call it an *insertion* (design-reduction) morphism, after Tjur [36, p. 1299]: his elementary criterion quantifies over removals of sample points, and on the quantities of §5 that quantification acts through precisely this class (Lemma 5.2), a proper subclass of the morphisms over which McCullagh’s naturality quantifies; §5 keeps the two apart. Every injection is a bijection followed by an insertion, so selections in general are relabellings composed with insertions.
 - ψ *bijective*: *relabelling* of the treatments.
 - ψ *constant*: collapse of all treatments to a single one, the design in which no treatment distinction is recorded.

Smaller choices of $\mathbf{\Omega}$ are legitimate and impose weaker naturality demands: $\mathbf{\Omega}_{\text{merge}} \subset \mathbf{\Omega}_{\text{lab}}$, the wide subcategory of surjections alone, and the two-object truncations $\mathbf{\Omega}_{\text{arr}} \subset \mathbf{\Omega}_{\text{full}}$ used in Example 6.6. Since naturality over a category restricts to naturality over any subcategory, the verification below covers all of them at once.

3. **Response scales.** $\mathbf{V}$ has the single object $\mathbb{R}$ with its Borel σ -algebra and only the identity morphism, so that the response-scale naturality condition (displayed as (3) in §2.2) is vacuous. Affine equivariance of the one-way layout is true, but it is not an instance of (3): see Remark 2.12.

4. **Sample spaces.** $\mathbf{S} = \mathbf{V} \times \mathbf{U}^{\text{op}}$ and $\Gamma(\mathbb{R}, u) = \mathbb{R}^u$ with the product σ -algebra, the space of real response vectors $y = (y_i)_{i \in u}$. On $\varphi^{\text{op}}: u' \rightarrow u$ in $\mathbf{U}^{\text{op}}$, Γ gives restriction $\text{res}_\varphi: \mathbb{R}^{u'} \rightarrow \mathbb{R}^u$, $y \mapsto y \circ \varphi$.
5. **Designs.** $\mathbf{C} = \mathbf{Set}$ and both J 's are the underlying-set functors, so an object of $\mathbf{D}$ is a triple (u, ω, x) with $x: u \rightarrow \omega$ the *treatment assignment*: $x(i)$ is the treatment received by unit i . Equivalently, x is the partition of u into *groups*

$$u_a := x^{-1}(a) \subseteq u, \quad a \in \omega, \quad n_a := |u_a|, \quad \sum_{a \in \omega} n_a = n,$$

indexed by ω and allowed to have empty parts. The design is *balanced* when all n_a are equal, and x is surjective exactly when every treatment is observed. A morphism $(\varphi, \psi): (u, \omega, x) \rightarrow (u', \omega', x')$ is a subsample inclusion together with a treatment map such that

$$x'(\varphi(i)) = \psi(x(i)) \quad (i \in u),$$

the comma condition, displayed as (1) in §2.2: *each retained unit carries, in the larger design, the ψ -image of the treatment it carried in the smaller one.*

6. **Parameters.** $\mathbf{K}$ has objects $W \times \mathbb{R}_{>0}$ with $W \in \mathbf{Vect}_{\mathbb{R}}^{\text{fd}}$, morphisms $A \times \text{id}_{\mathbb{R}_{>0}}$ with A linear, and $U_{\mathbf{K}}$ the underlying set, faithful (the same parameter category is used in Example 2.8 below). The parameter functor is

$$\Theta(\omega) = \mathbb{R}^\omega \times \mathbb{R}_{>0}, \quad \Theta(\psi)(\mu', \sigma^2) = (\mu' \circ \psi, \sigma^2), \quad \text{i.e.} \quad (\Theta(\psi)\mu')_a = \mu'_{\psi(a)}.$$

A parameter value is a pair $\theta = (\mu, \sigma^2)$ with $\mu = (\mu_a)_{a \in \omega}$ the vector of *group means*, one real number per treatment label, not per unit, and σ^2 a single *common* (homoscedastic) error variance, shared by all groups and therefore untouched by $\Theta(\psi)$. Precomposition ψ^* is linear, so Θ does land in $\mathbf{K}$, and it is contravariant because a map of label sets pulls means back, not forward.

The model. For $d = (u, \omega, x)$ and $\theta = (\mu, \sigma^2) \in \mathbb{R}^\omega \times \mathbb{R}_{>0}$,

$$P_{\mathbb{R}, d}(\cdot \mid \mu, \sigma^2) = \bigotimes_{i \in u} \mathcal{N}(\mu_{x(i)}, \sigma^2) \quad \text{on } \mathbb{R}^u,$$

that is, $Y_i = \mu_{x(i)} + \varepsilon_i$ with ε_i independent $\mathcal{N}(0, \sigma^2)$: the textbook Gaussian one-way layout, with group sizes n_a read off the design and no constraint (such as $\sum_a \mu_a = 0$) imposed on μ .

The example is carried through the whole paper: as a family of Markov kernels in Example 4.6, as the test case for meaningfulness in §5, and, after truncating Ω to two objects and discretising Θ , as the finite Baues–Wirsching computation of Example 6.6.

2.1 Brøns' comma-category equivalence

Brøns' key structural observation is that the somewhat unwieldy Definition 2.3 is equivalent to a diagram in a single fixed category of statistical models. Recall that for a functor $F: \mathbf{B} \rightarrow \mathbf{E}$, the comma category $(\text{id}_{\mathbf{E}} \downarrow F)$ has objects the triples (e, b, p) with $p: e \rightarrow Fb$ in $\mathbf{E}$.

Definition 2.6. The *category of statistical models* is

$$\text{StatMod} := (\text{id}_{\mathbf{Set}} \downarrow \mathcal{P})$$

formed from $\mathcal{P}: \mathbf{Meas} \rightarrow \mathbf{Set}$: an object is a set Θ_0 , a measurable space S , and a map $\Theta_0 \rightarrow \mathcal{P}(S)$, i.e. exactly a classical parametrised family of probability measures; a morphism is a pair (reparametrisation, measurable map of sample spaces) making the square commute.

Proposition 2.7 (after Brøns [9, pp. 1280, 1282]). *There is a bijection, natural in all the data, between*

1. natural transformations $P: L \Rightarrow R$ as in Definition 2.3, and
2. functors $\widehat{P}: \mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \mathbf{StatMod}$ whose composites with the two projections of the comma category are the given functors $U_K \Theta \text{pr}_\Omega^{\text{op}} \text{pr}_2$ (to $\mathbf{Set}$) and $\Gamma(\text{id} \times \text{pr}_\mathbf{U}^{\text{op}})$ (to $\mathbf{Meas}$).

Proof. This is an instance of the universal property of the comma category: for functors $G: \mathbf{A} \rightarrow \mathbf{E}$, $H: \mathbf{A} \rightarrow \mathbf{B}$ and $F: \mathbf{B} \rightarrow \mathbf{E}$, lifts of (G, H) to $\mathbf{A} \rightarrow (\text{id}_\mathbf{E} \downarrow F)$ commuting with both projections correspond bijectively to natural transformations $G \Rightarrow FH$. Apply this with $\mathbf{A} = \mathbf{V} \times \mathbf{D}^{\text{op}}$, $\mathbf{E} = \mathbf{Set}$, $\mathbf{B} = \mathbf{Meas}$, $G = L$, $H = \Gamma(\text{id} \times \text{pr}_\mathbf{U}^{\text{op}})$, $F = \mathcal{P}$: an object assignment $a \mapsto (Ga, Ha, P_a)$ is functorial precisely when each P -square commutes, which is naturality of P ; morphism-level data of $\widehat{P}$ are forced, so the correspondence is bijective. Naturality in the data is a routine check. $\square$

Brøns states the result with a sketch; the proof given above is complete.

2.2 Two worked instantiations of the design data

Definition 2.1 asks for six items, and it is easy to lose sight of how ordinary the resulting objects are. We therefore instantiate it twice, for multiple linear regression and for binary classification, verify the naturality conditions of Definition 2.3 in each case, and record two non-examples. Throughout, $\mathbf{FinSet}_{\text{inj}}$ denotes finite sets and injections and $\mathbf{Vect}_\mathbb{R}^{\text{fd}}$ finite-dimensional real vector spaces and linear maps.

item of Def. 2.1	A: linear regression	B: binary classification
(1) $\mathbf{U}$	$\mathbf{FinSet}_{\text{inj}}$	$\mathbf{FinSet}_{\text{inj}}$
(2) $\mathbf{\Omega}$	$\mathbf{Vect}_\mathbb{R}^{\text{fd}}$	$\mathbf{Vect}_\mathbb{R}^{\text{fd}}$
(3) $\mathbf{V}$	$\{\mathbb{R}\}$, identities only	$\{\mathbf{2}, \mathbf{1}\}$, with $!: \mathbf{2} \rightarrow \mathbf{1}$
(4) $\Gamma(v, u)$	$\mathbb{R}^u$, Borel product	v'' , discrete
(5) $\mathbf{D}$	$(J_\mathbf{U} \downarrow J_\mathbf{\Omega})$: design matrices	idem: feature matrices
(6) $\Theta(\omega)$	$\omega^* \times \mathbb{R}_{>0}$	ω^*

In both cases $\mathbf{C} = \mathbf{Set}$ and $J_\mathbf{U}, J_\mathbf{\Omega}$ are the underlying-set functors, so an object of $\mathbf{D}$ is a triple (u, ω, x) with $x: u \rightarrow \omega$ a set map, that is, exactly a design or feature matrix with rows indexed by units, and a morphism $(u, \omega, x) \rightarrow (u', \omega', x')$ is a pair (φ, ψ) with

$$x' \circ \varphi = \psi \circ x. \quad (1)$$

Condition (1), together with the injectivity of φ built into $\mathbf{U} = \mathbf{FinSet}_{\text{inj}}$ and the finiteness of u , is all that the verifications below consume (injectivity is used at the marginalisation step, and is necessary: for a non-injective φ square (2) fails in all three models, two units with the same image becoming identical rather than independent). This is worth emphasising: the comma-category definition of $\mathbf{D}$ is not bookkeeping; it is precisely the compatibility a model needs.

Fix $v \in \mathbf{V}$ and let $f = (\varphi, \psi): d = (u, \omega, x) \rightarrow d' = (u', \omega', x')$ in $\mathbf{D}$. In $\mathbf{D}^{\text{op}}$ this is an arrow $d' \rightarrow d$, and the square required by Definition 2.3 is

$$\begin{array}{ccc}
 U_K \Theta(\omega') & \xrightarrow{P_{v,d'}} & \mathcal{P}(\Gamma(v, u')) \\
 \Theta(\psi) \downarrow & & \downarrow \mathcal{P}(\text{res}_\varphi) \\
 U_K \Theta(\omega) & \xrightarrow{P_{v,d}} & \mathcal{P}(\Gamma(v, u))
 \end{array} \quad (2)$$

where $\text{res}_\varphi: v'' \rightarrow v''$, $y \mapsto y \circ \varphi$, is restriction along φ . In words: *marginalise the big model onto the subsample and one obtains the small model at the pulled-back parameter*. Separately, for $\alpha: v \rightarrow v'$ in $\mathbf{V}$ at fixed d , since L does not depend on the $\mathbf{V}$ coordinate, the square collapses to

$$\mathcal{P}(\alpha'') \circ P_{v,d} = P_{v',d} \quad \text{on } U_K \Theta(\omega). \quad (3)$$

A. Multiple linear regression

Example 2.8 (Gaussian linear model).

1. **Units.** $\mathbf{U} = \text{FinSet}_{\text{inj}}$; an object u indexes a sample, a morphism $\varphi: u \hookrightarrow u'$ is the inclusion of a subsample.
2. **Covariate spaces.** $\Omega = \text{Vect}_{\mathbb{R}}^{\text{fd}}$. An object $\omega \cong \mathbb{R}^p$ is the space in which covariate vectors live; a morphism $\psi: \omega \rightarrow \omega'$ is the inclusion of a submodel's covariates into a larger set, or a linear recoding (centring, orthogonalisation, contrast coding).
3. **Response scales.** $\mathbf{V}$ has the single object $\mathbb{R}$ (Borel) and only the identity morphism; see Remark 2.12 for why the affine group cannot be used here without amending Definition 2.1.
4. **Sample spaces.** $\mathbf{S} = \mathbf{V} \times \mathbf{U}^{\text{op}}$ and $\Gamma(\mathbb{R}, u) = \mathbb{R}^u$ with the product σ -algebra; on $\varphi^{\text{op}}: u' \rightarrow u$ in $\mathbf{U}^{\text{op}}$, Γ gives restriction $\text{res}_{\varphi}: \mathbb{R}^{u'} \rightarrow \mathbb{R}^u$.
5. **Designs.** $\mathbf{D} = (J_{\mathbf{U}} \downarrow J_{\Omega})$; an object (u, ω, x) is a design matrix, $x(i) \in \omega$ being the covariate vector of unit i . Choosing a basis of $\omega \cong \mathbb{R}^p$ and an enumeration of u , the map x is the $n \times p$ matrix X .
6. **Parameters.** $\mathbf{K}$ has objects $W \times \mathbb{R}_{>0}$ for $W \in \text{Vect}_{\mathbb{R}}^{\text{fd}}$ and morphisms $A \times \text{id}_{\mathbb{R}_{>0}}$ with A linear; $U_{\mathbf{K}}$ is the underlying set, faithful because a linear map is determined by its underlying function. The parameter functor is the dual, the subrepresentation of linear functionals inside McCullagh's standard representation [30, §4.2, p. 1235],

$$\Theta(\omega) = \omega^* \times \mathbb{R}_{>0}, \quad \Theta(\psi)(\beta', \sigma^2) = (\beta' \circ \psi, \sigma^2).$$

The model is, for $d = (u, \omega, x)$ and $\theta = (\beta, \sigma^2) \in \omega^* \times \mathbb{R}_{>0}$,

$$P_{\mathbb{R},d}(\cdot \mid \beta, \sigma^2) = \bigotimes_{i \in u} \mathcal{N}(\beta(x(i)), \sigma^2) \quad \text{on } \mathbb{R}^u,$$

that is, $Y \sim \mathcal{N}_n(X\beta, \sigma^2 I_n)$ in coordinates.

Verification of (2). Take $f = (\varphi, \psi): d \rightarrow d'$ and $\theta' = (\beta', \sigma^2)$ on ω' . Down-then-across, $\Theta(\psi)\theta' = (\beta' \circ \psi, \sigma^2)$, so

$$P_{\mathbb{R},d}(\cdot \mid \Theta(\psi)\theta') = \bigotimes_{i \in u} \mathcal{N}(\beta'(\psi(x(i))), \sigma^2).$$

Across-then-down, $P_{\mathbb{R},d'}(\cdot \mid \theta') = \bigotimes_{j \in u'} \mathcal{N}(\beta'(x'(j)), \sigma^2)$, and pushing forward along res_{φ} is marginalisation onto the coordinates $\varphi(u) \subseteq u'$, which by independence gives $\bigotimes_{i \in u} \mathcal{N}(\beta'(x'(\varphi(i))), \sigma^2)$. The two agree because $x'(\varphi(i)) = \psi(x(i))$, which is (1). The argument uses exactly two structural facts: independence across units, so that marginalisation is computed coordinatewise, and the fact that the mean of unit i depends on the design only through $x(i)$. Condition (3) is vacuous because $\mathbf{V}$ is discrete. $\square$

Remark 2.9 (Two familiar statistical statements, recovered). Specialising (2): with $\psi = \text{id}_{\omega}$ and φ a proper inclusion, the square says the model is consistent under subsampling, the marginal of an n -observation regression on a subsample is the regression on that subsample. With $\varphi = \text{id}_u$ and $\psi: \omega \hookrightarrow \omega'$ the inclusion of a submodel's covariate space, it says that fitting the large model at a β' annihilating the extra directions agrees with fitting the submodel at $\beta'|_{\omega}$. Model nesting thus appears as a naturality condition rather than as a constraint on coefficients.

Remark 2.10 (Intercept). As stated, the model has no intercept. Two clean repairs: take Ω to be finite-dimensional real *affine* spaces with affine maps and $\Theta(\omega)$ the affine functionals on ω times $\mathbb{R}_{>0}$; or keep Vect and restrict $\mathbf{D}$ to designs whose covariate maps land in an affine hyperplane $\{x_0 = 1\}$, with Ω cut down to the linear maps preserving it. The first is the more natural and is what the regression examples of [30, §6] effectively use.

Remark 2.11 (Design-conditional identifiability). $\Theta(\psi) = \psi^* \times \text{id}$ is injective iff ψ is surjective, and $L(v, d)$ parametrises the design d faithfully only relative to the image of x . The design-conditional statement, that β is identified from d only on the span of $x(u) \subseteq \omega$, is invisible at the level of Definition 2.1; it is (2) restricted to the subcategory of designs of full column rank. This is the precise point at which design-conditional sufficiency and identifiability enter the framework.

Remark 2.12 (Response-scale morphisms express *invariance*, not *equivariance*). Because $L = U_K \Theta \text{pr}_\Omega^{\text{op}} \text{pr}_2$ has no V dependence, a non-identity $\alpha: v \rightarrow v'$ imposes (3), namely $\mathcal{P}(\alpha^u) \circ P_{v,d} = P_{v',d}$ at the same parameter value. For $\alpha(y) = ay + b$ the left-hand side is $\bigotimes_i \mathcal{N}(a\beta(x(i)) + b, a^2 \sigma^2)$, which is not $P_{\mathbb{R},d}(\cdot | \beta, \sigma^2)$ unless $(a, b) = (1, 0)$. The Gaussian linear model is therefore *not* natural over $V = \text{Aff}(\mathbb{R})$, and taking V discrete is the correct instantiation.

Affine equivariance is of course a true and important property of the model, but it is the statement that the model is a fixed point of an *action* on the parameter, which in the McCullagh–Brøns hexagon requires the parameter functor to be defined on $V^{\text{op}} \times \Omega^{\text{op}}$ rather than on Ω^{op} alone. That is a mild generalisation of Definition 2.1, not an instance of it, and, as recorded in the Remark following the hexagon in §2, it is a return to McCullagh’s own construction rather than a departure from it: his parameter is a subrepresentation of v^ω and so moves with the response scale, whereas the V -independence of Θ is an artefact of Brøns’ packaging. In particular Example 2.5 takes V discrete for exactly this reason: the affine equivariance of the one-way layout is a true statement about an action on Θ , and lies outside Definition 2.1 as written, whereas its equivariance under *relabelling of treatments* is genuine naturality, because that acts through Ω^{op} .

B. Binary classification

Example 2.13 (Bernoulli model with a linear predictor).

1. **Units.** $U = \text{FinSet}_{\text{inj}}$, as above.
2. **Covariate spaces.** $\Omega = \text{Vect}_{\mathbb{R}}^{\text{fd}}$; objects are feature spaces, morphisms are linear feature maps.
3. **Response scales.** V has two objects, $\mathbf{2} = \{0, 1\}$ and $\mathbf{1} = \{*\}$, and as morphisms the identities together with the unique map $!: \mathbf{2} \rightarrow \mathbf{1}$. Statistically: one may discard the label, not relabel it.
4. **Sample spaces.** $\Gamma(v, u) = v^u$ with the discrete σ -algebra, so $\Gamma(\mathbf{2}, u) = \{0, 1\}^u$ is the label space and $\Gamma(\mathbf{1}, u)$ a one-point space; on φ^{op} , restriction as before.
5. **Designs.** $D = (J_U \downarrow J_\Omega)$; an object (u, ω, x) is the feature matrix, the labels being absent. The labels are the random object and hence live in Γ , not in D , which is the categorical form of the statement that classification is a model conditional on the features.
6. **Parameters.** $K = \text{Vect}_{\mathbb{R}}^{\text{fd}}$ with underlying set U_K , and $\Theta(\omega) = \omega^*$, $\Theta(\psi)(\beta') = \beta' \circ \psi$: the weight covector pulls back along feature maps.

With $\text{expit}(t) = (1 + e^{-t})^{-1}$ the model is

$$P_{\mathbf{2},d}(\cdot | \beta) = \bigotimes_{i \in u} \text{Bern}(\text{expit}(\beta(x(i)))) \quad \text{on } \{0, 1\}^u, \quad P_{\mathbf{1},d}(\cdot | \beta) = \delta \text{ on } \{*\}^u.$$

Verification. For (2) at $v = \mathbf{2}$, marginalising a product of Bernoullis onto $\varphi(u)$ gives the first family below, and the pulled-back parameter gives the second,

$$\bigotimes_{i \in u} \text{Bern}(\text{expit}(\beta'(x'(\varphi(i))))), \quad \bigotimes_{i \in u} \text{Bern}(\text{expit}((\beta' \circ \psi)(x(i)))),$$

and these agree by (1). At $v = \mathbf{1}$ both sides are the unique measure on a one-point space. For (3) with $!: \mathbf{2} \rightarrow \mathbf{1}$, the pushforward of $P_{\mathbf{2},d}(\cdot | \beta)$ along $!^u: \{0, 1\}^u \rightarrow \{*\}^u$ is the Dirac measure, which is $P_{\mathbf{1},d}(\cdot | \beta)$ for every β . $\square$

Remark 2.14 (The link plays no role). The verification uses only that the inverse link is applied *pointwise to the linear predictor*: any measurable $g^{-1}: \mathbb{R} \rightarrow [0, 1]$ yields a natural family by the same argument. Naturality in $\mathbf{D}$ is thus the categorical content of the statement that a generalised linear model is determined by a link applied unit by unit, and conversely that any sample-level coupling of the predictions destroys it (Example 2.17).

Remark 2.15 (Why the label swap is excluded from $\mathbf{V}$). The involution $\tau(y) = 1 - y$ on $\mathbf{2}$ satisfies $\mathcal{P}(\tau'')P_{2,d}(\cdot | \beta) = \bigotimes_i \text{Bern}(\text{expit}(-\beta(x(i)))) = P_{2,d}(\cdot | -\beta)$, which is not $P_{2,d}(\cdot | \beta)$. By (3), this violates naturality. The correct reading is not that the model is defective, but that relabelling is an equivariance, the phenomenon of Remark 2.12 in its smallest possible instance, and hence a convenient test case for whether the $\mathbf{V}$ -dependent parameter functor is worth adopting.

Non-examples

Both families below are well-defined and design-indexed; neither is a McCullagh–Brøns model, and they fail in the same slot for the same reason.

Example 2.16 (Empirical centring of the covariates). On the data of Example 2.8 set

$$\tilde{P}_{\mathbb{R},d}(\cdot | \beta, \sigma^2) = \bigotimes_{i \in u} \mathcal{N}(\beta(x(i) - \bar{x}_d), \sigma^2), \quad \bar{x}_d = \frac{1}{|u|} \sum_{i \in u} x(i).$$

This fails (2) already for $\psi = \text{id}$ and φ a proper inclusion, since $\bar{x}_d \neq \bar{x}_{d'}$: the mean assigned to a given unit changes when other units are removed. The failure is in the $\mathbf{U}$ slot, and it is the mechanism behind Tjur’s design-averaged non-example (Example 5.5; cf. Remark 5.3). Note that $\bar{x}$ is a perfectly good statistic; what is illegitimate is its appearance in the *definition* of the family.

Example 2.17 (Sample-normalised classifier). On the data of Example 2.13 let

$$\tilde{P}_{2,d}(\cdot | \beta) = \bigotimes_{i \in u} \text{Bern}(s_i(\beta, d)), \quad s_i = \frac{e^{\beta(x(i))}}{\sum_{j \in u} e^{\beta(x(j))}},$$

a softmax taken across the sample rather than across the label set. Again the $\mathbf{U}$ slot fails: dropping a unit rescales every remaining probability. Normalisation across a batch is a routine operation in learning pipelines, so this is not a pathology invented for the purpose; its non-naturality is exactly the kind of obstruction that the cohomological formulation of §6 detects and classifies.

The three slots may now be summarised as follows.

slot	what naturality asserts	what it rules out
$\mathbf{U}$	marginal onto a subsample is the subsample model	sample-dependent normalisation, empirical centring
$\mathbf{\Omega}$	parameter pulls back along covariate maps	design-dependent reparametrisation
$\mathbf{V}$	<i>invariance</i> at fixed parameter	relabelling, rescaling (these are equivariances)

The verifications above are elementary. Remark 2.12 is the one substantive point: with Θ defined on Ω^{op} alone, response-scale morphisms can express only invariance, so textbook equivariance statements about linear models lie outside Definition 2.1 as written.

Remark 2.18 (The one-way layout is Example 2.8 in dummy coding). Write $\mathbf{D}_{\text{lab}}$ for the design category of Example 2.5 and $\mathbf{D}_{\text{lin}}$ for that of Example 2.8. Throughout this remark, ω denotes an object of Ω_{lab} , a finite treatment-label set as in Example 2.5, carrying no linear structure, *not* the covariate space $\omega \cong \mathbb{R}^p$ of Example 2.8, which appears here as $\mathbb{R}[\omega]$. Let $\mathbb{R}[-]: \mathbf{FinSet} \rightarrow \mathbf{Vect}_{\mathbb{R}}^{\text{fd}}$ be the free-vector-space functor and $\eta_{\omega}: \omega \rightarrow \mathbb{R}[\omega]$ the basis inclusion, one-hot coding of a treatment label, which is natural, $\mathbb{R}[\psi] \eta_{\omega} = \eta_{\omega'} \psi$. Then

$$E(u, \omega, x) = (u, \mathbb{R}[\omega], \eta_{\omega} \circ x), \quad E(\varphi, \psi) = (\varphi, \mathbb{R}[\psi])$$

is a functor $D_{\text{lab}} \rightarrow D_{\text{lin}}$: it is well defined on morphisms because $\mathbb{R}[\psi]\eta_{\omega}x = \eta_{\omega'}\psi x = \eta_{\omega'}x'\varphi$, so (1) is transported. Under the canonical isomorphism $\mathbb{R}[\omega]^* \cong \mathbb{R}^{\omega}$, natural in ω , one has $\Theta_{\text{lab}} \cong \Theta_{\text{lin}} \circ \mathbb{R}[-]^{\text{op}}$ and $\beta(\eta_{\omega}x(i)) = \mu_{x(i)}$, whence

$$P_{\mathbb{R},d}^{\text{one-way}} = P_{\mathbb{R},Ed}^{\text{lin}} \quad \text{for every } d \in D_{\text{lab}}.$$

The one-way layout is therefore the reindexing of the Gaussian linear model along E , and its naturality is inherited, a natural transformation restricting along any functor. Two things are *not* inherited. First, the converse fails, for two distinct reasons. E is faithful but not full: a morphism $(\varphi, \ell): E(d) \rightarrow E(d')$ of D_{lin} must send $\eta_{\omega}(x(i))$ to $\eta_{\omega'}(x'(\varphi(i)))$ for every unit i , by (1), which pins ℓ down to some $\mathbb{R}[\psi]$ on the levels actually observed in d but leaves it an arbitrary linear map on the unobserved ones (when x is surjective, every morphism from $E(d)$ to an object in the image of E is itself in the image of E). And D_{lin} has objects outside the image of E altogether, recoded designs, whose covariate vectors are contrasts or sum-to-zero codings rather than indicators, together with the morphisms into and among them that no treatment map induces; centring, being affine rather than linear, is not a morphism of D_{lin} at all. So naturality over D_{lin} is strictly stronger than naturality over D_{lab} , and choosing label sets as Ω is a substantive weakening of the demand, not a change of notation. Second, the cohomological test case of §6 is computed over label morphisms, so it certifies obstructions relative to that smaller class only; this is one instance of the class-relativity insisted on in §6.

Example 2.5, continued: verification of the one-way layout. With the naturality squares (2)–(3) now displayed, we verify that the family of Example 2.5 is a McCullagh–Brøns model, and record what each condition says for it.

Verification of (2). Let $f = (\varphi, \psi): d \rightarrow d'$ and $\theta' = (\mu', \sigma^2)$ on ω' . Down-then-across, $\Theta(\psi)\theta' = (\mu' \circ \psi, \sigma^2)$, so

$$P_{\mathbb{R},d}(\cdot \mid \Theta(\psi)\theta') = \bigotimes_{i \in u} \mathcal{N}(\mu'_{\psi(x(i))}, \sigma^2).$$

Across-then-down, $P_{\mathbb{R},d'}(\cdot \mid \theta') = \bigotimes_{j \in u'} \mathcal{N}(\mu'_{x'(j)}, \sigma^2)$, and pushing forward along res_{φ} is marginalisation onto the coordinates $\varphi(u) \subseteq u'$, which by independence gives $\bigotimes_{i \in u} \mathcal{N}(\mu'_{x'(\varphi(i))}, \sigma^2)$. The two agree because $x'(\varphi(i)) = \psi(x(i))$, which is (1). As in Example 2.8 the argument consumes exactly two structural facts: independence across units, so that marginalisation is coordinatewise, and the fact that the law of unit i depends on the design only through $x(i)$. Condition (3) is vacuous because V is discrete. $\square$

What naturality says, slot by slot. Specialising the naturality conditions (2) and (3) to morphisms living in a single slot, and writing (N-C) for the resulting condition on morphisms of $C \in \{U, \Omega, V\}$:

(N-U) $\psi = \text{id}_{\omega}$, φ a proper inclusion: the marginal of the layout on the subsample $\varphi(u)$ is the layout on that subsample, at the same (μ, σ^2) and with the restricted assignment $x' \circ \varphi$. *Consistency under subsampling.*

(N- Ω , merge) $\varphi = \text{id}_u$, ψ surjective: the pulled-back mean vector $\mu' \circ \psi$ is constant on the blocks of ψ , and the square says that the fine-grained model evaluated at such a block-constant mean vector *is* the merged model at μ' . Naturality does not assert that merged groups have equal means; it asserts that on the subfamily where they do, the two descriptions of the experiment coincide.

(N- Ω , select) ψ injective: the same square read on a subfamily of treatments. For ψ an inclusion $\omega \subseteq \omega'$ this is design reduction, the covariate shadow of Tjur’s sample-point removals, and by Lemma 5.2 the only part of (N- Ω , select) that his criterion reaches; a general injection adds a relabelling of the surviving groups, which his criterion does not see. The distinction is not idle: in Example 6.7(i) the merge morphisms alone certify nothing, and it is the selection morphisms that detect the failure.

(N-V) Vacuous here: by item 3 of Example 2.5, V has only identity morphisms, so (3) imposes nothing.

Automorphisms: exchangeability and label equivariance. Taking $d' = d$ with φ, ψ bijections, (1) forces φ to carry u_a onto $u_{\psi(a)}$, so

$$\text{Aut}_{\mathbb{D}}(u, \omega, x) = \{(\varphi, \psi) : \psi \in \mathfrak{S}_{\omega}, n_{\psi(a)} = n_a \forall a, \varphi|_{u_a} : u_a \xrightarrow{\sim} u_{\psi(a)}\},$$

an extension of the group of size-preserving relabellings by $\prod_{a \in \omega} \mathfrak{S}_{n_a}$, split once enumerations of the groups are chosen. Naturality at (φ, id) is invariance of $P_{\mathbb{R}, d}(\cdot \mid \theta)$ under permuting coordinates within groups, i.e. *within-group exchangeability*; naturality at a general (φ, ψ) is $(\text{res}_{\varphi})_* P_{\mathbb{R}, d}(\cdot \mid \mu, \sigma^2) = P_{\mathbb{R}, d}(\cdot \mid \mu \circ \psi, \sigma^2)$, i.e. *equivariance* under relabelling of treatments. That equivariance is available in the Ω slot but not in the V slot is not an accident of this example: Θ is defined on Ω^{op} , so a covariate morphism can act on the parameter, whereas a response-scale morphism cannot (Remark 2.12).

Identifiability and design-conditional sufficiency. $\Theta(\psi) = \psi^* \times \text{id}$ is injective iff ψ is surjective and surjective iff ψ is injective. Independently of that, μ_a enters $P_{\mathbb{R}, d}$ only for $a \in x(u)$: at a design that leaves some treatment unobserved, $P_{\mathbb{R}, d}$ factors through the restriction $\mathbb{R}^{\omega} \rightarrow \mathbb{R}^{x(u)}$ and μ is identified only on the observed levels, the exact analogue, for label sets, of the column-rank condition of Remark 2.11. The design-conditional sufficient statistic is the pair (group sample means on $x(u)$, pooled within-group sum of squares), and it is at this point that design-conditional sufficiency and identifiability enter.

3 Markov-categorical background

We use only a small fragment of the theory; see [14, 32] for the general framework.

The *Giry monad* $(\mathcal{G}, \eta, \mu)$ is the probability-distribution monad on **Meas**.

For a measurable space $X = (X, \Sigma_X)$, $\mathcal{G}X$ is the set of all probability measures on X . This set is itself equipped with a σ -algebra: namely, the smallest σ -algebra making all evaluation maps

$$\text{ev}_A : \mathcal{G}X \rightarrow [0, 1], \quad \text{ev}_A(\mu) = \mu(A),$$

measurable, for every $A \in \Sigma_X$.

Thus $\mathcal{G}X$ is again a measurable space.

If

$$f : X \rightarrow Y$$

is measurable, define

$$\mathcal{G}f : \mathcal{G}X \rightarrow \mathcal{G}Y$$

by pushforward:

$$\mathcal{G}f(\pi) = f_*\pi,$$

that is,

$$(f_*\pi)(B) = \pi(f^{-1}(B)), \quad B \in \Sigma_Y.$$

Hence

$$\mathcal{G} : \mathbf{Meas} \rightarrow \mathbf{Meas}$$

is a functor.

The unit of the monad is

$$\eta_X : X \rightarrow \mathcal{G}X, \quad \eta_X(x) = \delta_x,$$

where δ_x denotes the Dirac probability measure concentrated at x .

The multiplication of the monad is

$$\mu_X : \mathcal{G}\mathcal{G}X \rightarrow \mathcal{G}X,$$

defined by

$$\mu_X(\Xi)(A) = \int_{\mathcal{G}X} \nu(A) d\Xi(\nu), \quad A \in \Sigma_X.$$

Thus an element $\Xi \in \mathcal{GG}X$, i.e. a probability distribution on the space of probability distributions on X , is sent to its mixture.

Equivalently, for every bounded measurable function $h : X \rightarrow \mathbb{R}$,

$$\int_X h(x) d\mu_X(\Xi)(x) = \int_{\mathcal{GX}} \left(\int_X h(x) d\nu(x) \right) d\Xi(\nu).$$

The triple

$$(\mathcal{G}, \eta, \mu)$$

satisfies the monad laws.

A Kleisli morphism for the Giry monad from X to Y is a measurable map

$$K : X \rightarrow \mathcal{GY}.$$

Equivalently, it is a Markov kernel

$$K : X \times \Sigma_Y \rightarrow [0, 1]$$

such that, for every $x \in X$,

$$B \mapsto K(x, B)$$

is a probability measure on Y , and for every $B \in \Sigma_Y$,

$$x \mapsto K(x, B)$$

is measurable.

Given two Kleisli morphisms

$$K : X \rightarrow \mathcal{GY}, \quad L : Y \rightarrow \mathcal{GZ},$$

their Kleisli composition is

$$L \odot K : X \rightarrow \mathcal{GZ}$$

defined by

$$(L \odot K)(x)(C) = \int_Y L(y)(C) dK(x)(y), \quad C \in \Sigma_Z.$$

In other words, the Kleisli category of the Giry monad has measurable spaces as objects and Markov kernels as morphisms. This leads to the following definition.

Definition 3.1. The Kleisli category of the Giry monad is called **Stoch**, and it is the category whose objects are measurable spaces and whose morphisms $f : X \rightsquigarrow Y$ are Markov kernels $f : X \times \Sigma_Y \rightarrow [0, 1]$, composed by the Chapman–Kolmogorov formula, see [14, 20, 32]. There is a natural bijection

$$\text{Stoch}(X, Y) \cong \text{Meas}(X, \mathcal{GY}). \quad (4)$$

FinStoch is the full subcategory of finite discrete measurable spaces; morphisms are stochastic matrices.

Stoch is a Markov category: symmetric monoidal under the product of measurable spaces, with copy and discard maps at every object, naturality being imposed on discard alone [14, Def. 2.1]; we write $\text{copy}_X : X \rightarrow X \otimes X$ for the copy map at X .

A morphism $f : X \rightarrow Y$ of a Markov category is *deterministic* if it is a homomorphism of the copy comonoids, that is, if

$$\text{copy}_Y \circ f = (f \otimes f) \circ \text{copy}_X$$

[14, Def. 10.1]. Read left to right, the condition says that running f once and duplicating its output agrees with duplicating the input and running f twice independently; any genuine randomness in f breaks this, since two independent runs need not agree. In **Stoch** the condition says exactly that $f(x, B) \in \{0, 1\}$ for every $x \in X$ and every $B \in \Sigma_Y$, i.e. that each $f(x, \cdot)$ is a $\{0, 1\}$ -valued measure [14, Ex. 10.4]; in **FinStoch**, that the stochastic matrix of f has entries in $\{0, 1\}$. Deterministic morphisms contain the identities and are closed under composition and under $\otimes$, so they form a wide symmetric monoidal subcategory $\text{Stoch}_{\text{det}} \subseteq \text{Stoch}$.

Definition 3.2. The *Dirac embedding* $\delta: \mathbf{Meas} \rightarrow \mathbf{Stoch}$ is the identity on objects and sends a measurable map f to the kernel $x \mapsto \delta_{f(x)}$. It is strong monoidal and lands in $\mathbf{Stoch}_{\text{det}}$, since a Dirac kernel is $\{0, 1\}$ -valued; it is faithful on the measurable maps into Y exactly when Σ_Y separates points (so on all standard Borel, discrete and finite codomains, but not on $\mathbf{Meas}$ as a whole). The converse, that every deterministic $X \rightsquigarrow Y$ is δ of a measurable map, is not automatic: it asks that the $\{0, 1\}$ -valued measures on Y be exactly its points. This holds when Σ_Y is countably generated and separates points, in particular for standard Borel Y , and always in $\mathbf{FinStoch}$ [14, Ex. 10.4]. Hypotheses (a) and (b) of Proposition 4.11 are the two instances of this requirement that arise below; see Remark 4.13 (ii).

Two facts are used below without further comment: (i) correspondence (4) is natural in X (via precomposition with measurable maps) and in Y (via pushforward); (ii) a diagram of Dirac kernels δf_j commutes in $\mathbf{Stoch}$ iff the diagram of the f_j commutes in $\mathbf{Meas}$, provided the codomains involved separate points, since δ is faithful.

Remark 3.3 (What Markov categories add). The point of the change of codomain in Section 4 is not notational. Once a model lands in $\mathbf{Stoch}$, the synthetic apparatus of [14], a.s.-equality and the synthetic Fisher–Neyman characterisation of sufficiency (which requires strict positivity, available in $\mathbf{Stoch}$ [14, Ex. 13.19, Thm. 14.5], not conditionals, which $\mathbf{Stoch}$ lacks in general [14, Ex. 11.3]), applies to each component $P_{v,d}$ inside the category, while McCullagh–Brøns naturality constrains the family *across* components. The two mechanisms compose: e.g. sufficiency of a statistic for every design, plus naturality of the statistic, is a statement with actual content about the whole diagram, not a componentwise remark. Making such combined statements provable is the purpose of the bridge.

4 The bridge

The observation driving everything is: by (4), a component $P_{v,d}: U_K \Theta(\omega) \rightarrow \mathcal{P}(\Gamma(v, u))$ of a McCullagh–Brøns model *wants* to be a morphism $\Theta(\omega) \rightsquigarrow \Gamma(v, u)$ in $\mathbf{Stoch}$, the only gap between Definition 2.3 and that statement is measurability in θ , which the $\mathbf{Set}$ -valued formulation cannot see. In the finite case the gap vanishes, and the translation is an equivalence; in general it is an added hypothesis, which we make explicit.

4.1 Finite case

Assume the data of Definition 2.1 are *finite*: $\mathbf{V}$, $\mathbf{\Omega}$, $\mathbf{U}$ take values with finite underlying sets, Γ lands in finite discrete spaces, and $U_K \Theta(\omega)$ is finite for every ω . Equip each $U_K \Theta(\omega)$ with its discrete measurable structure, so that $U_K \Theta$ lifts (uniquely) to a functor $\Theta_{\mathbf{Meas}}: \mathbf{\Omega}^{\text{op}} \rightarrow \mathbf{Meas}$ with $U_K \Theta = |-| \circ \Theta_{\mathbf{Meas}}$, where $|-|: \mathbf{Meas} \rightarrow \mathbf{Set}$ is the forgetful functor.

Define two functors into $\mathbf{FinStoch}$:

$$\mathbf{L} := \delta \circ \Theta_{\mathbf{Meas}} \circ \text{pr}_{\mathbf{\Omega}}^{\text{op}} \circ \text{pr}_2, \quad \mathbf{R} := \delta \circ \Gamma \circ (\text{id}_{\mathbf{V}} \times \text{pr}_{\mathbf{U}}^{\text{op}}),$$

both $\mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \mathbf{FinStoch}$, where δ is the Dirac embedding of Definition 3.2.

Theorem 4.1 (Bridge, finite case). *There is a canonical bijection between:*

1. *McCullagh–Brøns models* $P: \mathbf{L} \Rightarrow \mathbf{R}$ (Definition 2.3);
2. *families of morphisms* $\mathbf{P}_{v,d}: \mathbf{L}(v, d) \rightsquigarrow \mathbf{R}(v, d)$ in $\mathbf{FinStoch}$, indexed by the objects of $\mathbf{V} \times \mathbf{D}^{\text{op}}$, such that for every morphism m of $\mathbf{V} \times \mathbf{D}^{\text{op}}$ the square

$$\begin{array}{ccc} \mathbf{L}(v, d) & \rightsquigarrow^{\mathbf{P}_{v,d}} & \mathbf{R}(v, d) \\ \mathbf{L}(m) \downarrow & & \downarrow \mathbf{R}(m) \\ \mathbf{L}(v', d') & \rightsquigarrow^{\mathbf{P}_{v',d'}} & \mathbf{R}(v', d') \end{array} \quad (5)$$

commutes in $\mathbf{FinStoch}$; i.e. natural transformations $\mathbf{P}: \mathbf{L} \Rightarrow \mathbf{R}$.

Remark 4.2 (Where the probability sits in the square). In (2), the vertical arrows $\mathbf{L}(m) = \delta \Theta_{\text{Meas}}(\psi)$ and $\mathbf{R}(m) = \delta \Gamma(\alpha, \varphi)$ are Dirac kernels, hence deterministic, by the definition of $\mathbf{L}$ and $\mathbf{R}$; only the components $\mathbf{P}_{v,d}$ carry probability, and these are arbitrary stochastic matrices (Dirac only when the model is degenerate).

Proof of Theorem 4.1. Write $d = (u, \omega, x)$, so that $\mathbf{L}(v, d) = \delta \Theta_{\text{Meas}}(\omega)$ and $\mathbf{R}(v, d) = \delta \Gamma(v, u)$; since δ is the identity on objects, these are the measurable spaces $\Theta_{\text{Meas}}(\omega)$ and $\Gamma(v, u)$, both finite and discrete.

Step 1 (object level: transposition). Fix (v, d) . We claim a bijection

$$\text{Set}(U_K \Theta(\omega), \mathcal{P}(\Gamma(v, u))) \xrightarrow{\cong} \text{FinStoch}(\Theta_{\text{Meas}}(\omega), \Gamma(v, u)), \quad P_{v,d} \mapsto \mathbf{P}_{v,d}. \quad (6)$$

Indeed $U_K \Theta(\omega)$ is by construction the underlying set of the discrete space $\Theta_{\text{Meas}}(\omega)$, and $\mathcal{P}(S)$ is the underlying set of the measurable space $\mathcal{G}S$. Because $\Theta_{\text{Meas}}(\omega)$ is discrete, *every* function out of it is measurable, so $\text{Set}(U_K \Theta(\omega), \mathcal{P}(\Gamma(v, u))) = \text{Meas}(\Theta_{\text{Meas}}(\omega), \mathcal{G}\Gamma(v, u))$, an equality of sets, not merely an inclusion. This is the component of the adjunction $D \dashv U$ of Proposition 4.14 at $(U_K \Theta(\omega), \mathcal{G}\Gamma(v, u))$; what is done here by hand at one pair of objects is done there uniformly. This is the only place where discreteness of the parameter object is used, finiteness of the spaces serves only to place $\mathbf{L}$, $\mathbf{R}$ and $\mathbf{P}$ in FinStoch rather than in Stoch , and it is exactly the gap identified before Definition 2.3: in general one must *require* measurability in θ . Composing with (4) gives (6). Explicitly, $\mathbf{P}_{v,d}$ is the kernel $\theta \mapsto P_{v,d}(\cdot \mid \theta)$, the same data read as a morphism rather than as a $\mathcal{P}$ -valued function.

Step 2 (morphism level: both sides are deterministic). Let $m = (\alpha, (\varphi, \psi)^{\text{op}})$ be a morphism of $\mathbf{V} \times \mathbf{D}^{\text{op}}$, with target (v', d') , $d' = (u', \omega', x')$, so that $(\varphi, \psi): d' \rightarrow d$ in $\mathbf{D}$, with $\varphi: u' \rightarrow u$ and $\psi: \omega' \rightarrow \omega$. On the Set side $L(m) = U_K \Theta(\psi)$ and $R(m) = \mathcal{P}(\Gamma(\alpha, \varphi))$, the latter being pushforward of measures along $\Gamma(\alpha, \varphi)$. On the FinStoch side, by definition of $\mathbf{L}$ and $\mathbf{R}$,

$$\mathbf{L}(m) = \delta(\Theta_{\text{Meas}}(\psi)), \quad \mathbf{R}(m) = \delta(\Gamma(\alpha, \varphi)),$$

so the vertical arrows of (5) are Dirac kernels: $\mathbf{L}$ and $\mathbf{R}$ send every morphism to a deterministic kernel, while nothing constrains $\mathbf{P}_{v,d}$ itself.

Step 3 (the two transposition rules). Recall that composition in Stoch is given by $(k' \circ k)(x)(B) = \int k'(y)(B) k(x)(dy)$. Two special cases are all we need. For a measurable $g: Y \rightarrow Y'$ and a kernel $k: X \rightsquigarrow Y$,

$$(\delta(g) \circ k)(x)(B) = \int \delta_{g(y)}(B) k(x)(dy) = k(x)(g^{-1}B) = (\mathcal{G}(g) \circ k)(x)(B), \quad (7)$$

so postcomposing with $\delta(g)$ in FinStoch is postcomposing with the pushforward $\mathcal{P}(g) = \mathcal{G}(g)$ under (6). Dually, for a measurable $h: X' \rightarrow X$,

$$(k \circ \delta(h))(x')(B) = \int k(x)(B) \delta_{h(x')}(dx) = k(h(x'))(B), \quad (8)$$

so precomposing with $\delta(h)$ is precomposing with h itself. These are the two naturality statements of (4) recorded after Definition 3.2, here written out in full.

Step 4 (squares correspond). The Set -naturality square for P at m is the equality of functions $U_K \Theta(\omega) \rightarrow \mathcal{P}(\Gamma(v', u'))$

$$\mathcal{P}(\Gamma(\alpha, \varphi)) \circ P_{v,d} = P_{v',d'} \circ U_K \Theta(\psi).$$

Transpose each side under (6). By (7) the left-hand side transposes to $\delta(\Gamma(\alpha, \varphi)) \circ \mathbf{P}_{v,d} = \mathbf{R}(m) \circ \mathbf{P}_{v,d}$; by (8) the right-hand side transposes to $\mathbf{P}_{v',d'} \circ \delta(\Theta_{\text{Meas}}(\psi)) = \mathbf{P}_{v',d'} \circ \mathbf{L}(m)$. Since (6) is a bijection of hom-sets, the two functions are equal if and only if the two kernels are equal, i.e. if and only if square (5) commutes.

Step 5 (conclusion). Steps 1 and 4 give, for each family, a family with the same index set and the same commuting squares, and the assignment $P_{v,d} \mapsto \mathbf{P}_{v,d}$ is invertible objectwise by (6), with the inverse governed by the same two rules read right to left. Hence natural transformations $P: L \Rightarrow R$ correspond bijectively to natural transformations $\mathbf{P}: \mathbf{L} \Rightarrow \mathbf{R}$. The final assertions are Step 2 together with the observation that $\mathbf{P}_{v,d}$ lies in the image of δ precisely when each $P_{v,d}(\cdot \mid \theta)$ is a Dirac measure, which is not required of a statistical model. $\square$

The proof is a Kleisli-transposition argument; its value is not depth but the licence it grants: every synthetic notion available in FinStoch applies to finite McCullagh–Brøns models verbatim.

Is finiteness a conceptual restriction? Rounding as a V -morphism. A natural objection to the finiteness hypothesis of Theorem 4.1 is that it costs nothing: responses are bounded by the range of any real instrument, covariate ranges are bounded for the same reason, and a real number recorded in a 64-bit floating-point representation takes one of finitely many values, so the data of any actual experiment are finite and Theorem 4.1 already covers everything that is ever computed. Half of this is correct and worth recording precisely; the other half is not, and the two halves are best kept apart.

What is correct. Fix $M > 0$ and a finite grid $F \subset [-M, M]$, for definiteness, the floating-point numbers representable in that range, and let $\rho: \mathbb{R} \rightarrow F$ be rounding, with values outside $[-M, M]$ clamped to the extreme grid points. Then ρ is Borel measurable and surjective, hence precisely a coarsening morphism of the kind item 3 (the response-scale category V) of Definition 2.1 admits. Enlarge the response-scale category accordingly: let V_F have the two objects $\mathbb{R}$ and F , the grid carrying the discrete σ -algebra, and, besides the identities, the single morphism $\rho: \mathbb{R} \rightarrow F$; this is again a response-scale category in the sense of Definition 2.1(3), and Γ extends canonically by $\Gamma(F, u) = F^u$. Setting

$$P_{F,d}(\cdot \mid \theta) := (\rho^u)_* P_{\mathbb{R},d}(\cdot \mid \theta)$$

makes (3) hold by construction, and every component now has finite codomain. Two things follow. First, quantisation is not external to the framework and not an approximation to the model: the discretised model *is* the image of the model under a morphism of the enlarged response-scale category V_F , obtained with no hypothesis and no error term, and recorded data always live in the image of such a morphism. Second, this is one of the rare non-identity response-scale morphisms genuinely available to the Gaussian models of §2.2: unlike affine rescaling, which acts on the parameter and therefore lies outside Definition 2.1 as written (Remark 2.12), rounding is compatible with the model *at the same parameter value*, which is exactly what (3) demands.

What is not correct. The argument does not reduce Theorem 4.3 to Theorem 4.1, for three independent reasons.

(a) *It discretises the sample side only.* The hypothesis of Theorem 4.1 also requires $U_K \Theta(\omega)$ to be finite. Rounding is a morphism of V and leaves the parameter functor Θ untouched, whatever it is; in every model of §2.2 the parameter objects are continua, $\mathbb{R}^\omega \times \mathbb{R}_{>0}$ in the one-way layout of Example 2.5, for instance, so $P_{F,d}$ is still a family indexed by a continuum: it is an object for Theorem 4.3, whose measurability hypothesis it inherits from $P_{\mathbb{R},d}$ rather than avoids.

(b) *Discretising the parameter as well does not in general yield an instance of Definition 2.1.* Where $\Theta(\psi)$ is mere reindexing, as in the one-way layout, a finite grid of parameter values is preserved and Theorem 4.1 applies; this is the setting of §6. Where $\Theta(\psi) = \psi^*$ uses the linear structure of the covariate category, as in Examples 2.8 and 2.13, it does not: a floating-point grid is not closed under arithmetic, addition is not associative on it and $\rho(a+b) \neq \rho(a) + \rho(b)$ in general, so F^p is not a vector space, the covariate category $\text{Vect}_{\mathbb{R}}^{\text{fd}}$ of Example 2.8 has no floating-point analogue, and $\Theta(\psi) = \psi^*$ would be functorial only up to rounding. Naturality is an *equational* condition: the squares would commute approximately, and no notion of approximate naturality is developed here. The computation of §6 is carried out in exact arithmetic for this reason. Formulating an ε -naturality with a quantitative obstruction is a reasonable question and would be the honest home for numerical claims; nothing in this draft supplies one.

(c) *The statistical content is not preserved by ρ_* .* The pushforward of a Gaussian family along ρ is not an exponential family, and the pair (group means, within-group sum of squares) is not sufficient for it. Statements about sufficiency and about exponential families are theorems about the continuum object; its finite images do not inherit them.

The correct relation is a limit, not a rounding. Let $(\rho_n: S \rightarrow F_n)_n$ be a refining sequence of finite quantisations whose cells generate the Borel σ -algebra of S . Since the cells form a generating π -system, a probability measure is determined by its pushforwards, so the canonical map $\mathcal{P}(S) \rightarrow \varprojlim_n \mathcal{P}(F_n)$ is injective: nothing studied here is invisible at finite level, and in this sense finiteness is indeed not a conceptual limitation. But the map is not a bijection for free: the projective limit $\hat{S} = \varprojlim_n F_n$ is a compact

zero-dimensional space into which S embeds, and $\lim_{\leftarrow n} \mathcal{P}(F_n) \cong \mathcal{P}(\hat{S})$ by a Kolmogorov-type extension theorem; $\mathcal{P}(S)$ is then the subset of measures carried by the image of S , which is Borel for standard Borel S but is proper in general (for the dyadic quantisation of $[0, 1]$ the consistent family concentrating on the cells $[\frac{1}{2} - 2^{-n}, \frac{1}{2}]$ has no preimage). It is at exactly that point that the regularity hypothesis of Theorem 4.3 reappears rather than disappears. The measurable case is thus the projective limit of the finite ones, not a special case of them, and cannot be dispensed with by quantising the data.

The injectivity of $\mathcal{P}(S) \rightarrow \lim_{\leftarrow n} \mathcal{P}(F_n)$ is elementary; the identification of the projective limit with $\mathcal{P}(\hat{S})$ is classical but is not used below.

4.2 Measurable case

Theorem 4.3 (Bridge, measurable case). *Suppose the Set-valued parameter functor $U_K\Theta: \Omega^{\text{op}} \rightarrow \text{Set}$ lifts to a functor $\Theta_{\text{Meas}}: \Omega^{\text{op}} \rightarrow \text{Meas}$ along the underlying-set functor $|-|: \text{Meas} \rightarrow \text{Set}$, that is, $|-| \circ \Theta_{\text{Meas}} = U_K \circ \Theta$ (i.e. each parameter object carries a measurable structure and the reparametrisation maps $\Theta(\psi)$ are measurable), and let $P: L \Rightarrow R$ be a McCullagh–Brøns model such that each component $P_{v,d}: U_K\Theta(\omega) \rightarrow \mathcal{P}(\Gamma(v, u))$ is measurable as a map $\Theta_{\text{Meas}}(\omega) \rightarrow \mathcal{G}\Gamma(v, u)$ (equivalently: $\theta \mapsto P_{v,d}(A \mid \theta)$ is measurable for every measurable A); call such a P a measurable model. Define $\mathbf{L}, \mathbf{R}: \mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \text{Stoch}$ by the same formulas as in the finite case, $\mathbf{L} = \delta \Theta_{\text{Meas}} \text{pr}_\Omega^{\text{op}} \text{pr}_2$ and $\mathbf{R} = \delta \Gamma(\text{id}_V \times \text{pr}_U^{\text{op}})$. Then the conclusion of Theorem 4.1 holds with FinStoch replaced by Stoch : there is a canonical bijection between measurable models and natural transformations $\mathbf{P}: \mathbf{L} \Rightarrow \mathbf{R}$; the arrows $\mathbf{L}(m)$, $\mathbf{R}(m)$ are Dirac kernels and hence deterministic (Definition 3.2), with no further hypothesis; and each component $\mathbf{P}_{v,d}$ is a kernel that need not be deterministic.*

Proof. As in the finite case, with Step 1 replaced by the measurability hypothesis: (4) and its naturality hold for arbitrary measurable spaces; measurability of $P_{v,d}$ is exactly what places it in the hom-set

$$\text{Meas}(\Theta_{\text{Meas}}(\omega), \mathcal{G}\Gamma(v, u))$$

that (4) transposes; and the square-transposition argument (Steps 3–4) uses only naturality of the Kleisli bijection, not finiteness. $\square$

Remark 4.4 (The hypotheses are a genuine restriction). McCullagh’s formulation is Set-valued: parameter objects are bare sets (or objects of an abstract concrete category $\mathbf{K}$), and no measurability in θ is imposed. Theorem 4.3 therefore does *not* say “every McCullagh–Brøns model lives in Stoch ”; it says the ones satisfying a mild regularity hypothesis do. Two mitigations. (i) Every statistical model used in practice, in particular every exponential-family model, and all examples in [30] with finite-dimensional parameter spaces, satisfies the hypothesis with the obvious Borel structures (the measure- and function-valued parameter spaces of [30, §8] would require a choice of σ -algebra, not pursued here); kernels $\theta \mapsto P_\theta$ with measurable dependence are the standard regularity assumption of decision theory. (ii) The hypothesis is not merely harmless but *clarifying*: the Set-valued formulation silently permits models to which no measure-theoretic statistical procedure (risk, Bayes, estimation) could be applied; the bridge makes the regularity requirement part of the type. The bridge should therefore be stated with this hypothesis, not without it.

Remark 4.5 (No claim of essential surjectivity onto synthetic models). The bridge exhibits McCullagh–Brøns models as certain diagrams in Stoch . It does not claim that every “statistical model” definable synthetically in an abstract Markov category arises this way, nor that Stoch is the only reasonable codomain (quasi-Borel spaces [22] or Borel spaces are alternatives with better categorical properties). The choice of Stoch is the minimal one supporting the classic examples.

Example 4.6 (One-way layout, continued). In Example 2.5, $\Theta(\omega) = \mathbb{R}^\omega \times \mathbb{R}_{>0}$ with its Borel structure; measurability of $(\mu, \sigma^2) \mapsto \bigotimes_i N(\mu_{x(i)}, \sigma^2)$ is classical. Theorem 4.3 applies, and the three naturality conditions listed in Example 2.5 become commuting squares of kernels.

4.3 The category of statistical models, reformulated

It is natural to ask whether Brøns' category of statistical models is identified with a category built canonically from Stoch : objects its morphisms, morphisms the commuting squares with deterministic sides. The following remark lets us say precisely how much of it is true.

Remark 4.7 (The bridge is a representability computation). The proof of Theorem 4.1 is elementary because it is, in disguise, the composite of two standard facts. Isolating them is worth the space: they explain what the finiteness hypothesis is doing, and they are what the rest of this section runs on.

(i) *The probability functor is representable in Stoch .* The monoidal unit I of Stoch is the one-point measurable space, and

$$\text{Stoch}(I, S) = \text{Meas}(1, \mathcal{G}S) = \mathcal{G}S = \mathcal{P}(S),$$

naturally in S ; that is, $\mathcal{P} \cong \text{Stoch}(I, -)$. A probability measure is a generalised element at the unit, which is the precise content of the standard slogan that probability measures are *states*.

(ii) *Discrete parameter sets are copowers of the unit.* The Dirac embedding δ (Definition 3.2) is the free functor of the Kleisli adjunction $\text{Meas} \rightleftarrows \text{Stoch}$, hence a left adjoint, hence preserves colimits. Coproducts in Meas are disjoint unions (with the disjoint-union σ -algebra), so for any set A ,

$$\delta(A, 2^A) \cong \coprod_A I =: A \cdot I.$$

Combining, for any set A and measurable space S ,

$$\text{Set}(A, \mathcal{P}(S)) \cong \text{Set}(A, \text{Stoch}(I, S)) \cong \text{Stoch}(A \cdot I, S), \quad (9)$$

which is exactly the instance of (4) used in the proof of Theorem 4.1.

Two consequences. First, determinism of the parameter side is not an extra observation about the finite case: it is the statement that the domain object is a copower of I and that $\mathbf{L}$ sends every morphism to the Dirac kernel of a map of the indexing sets, so $\mathbf{L}$ lands in the image of δ by construction. Second, the finiteness hypothesis is doing two separable jobs, it makes every set-map into $\mathcal{P}(S)$ a kernel (that is (9)), and it makes δ full and faithful onto deterministic morphisms (that is the converse recorded in Definition 3.2), and only the second is a genuine restriction, since (9) holds verbatim over arbitrary sets.

Proposition 2.7 admits the same reading one level up: the comma category $(\text{id}_{\text{Set}} \downarrow \mathcal{P})$ represents, as an object of $\mathbf{Cat}$ and up to the evident 2-categorical refinements, the assignment $A \mapsto \{(G, H, \alpha) \mid \alpha: G \Rightarrow PH\}$, and the universal property invoked in its proof is the corresponding representability argument.

Definition 4.8. Let $q: X \rightsquigarrow Y$ and $q': X' \rightsquigarrow Y'$ be morphisms of Stoch . A *square* from q to q' is a pair (a, b) of deterministic morphisms $a: X \rightsquigarrow X'$, $b: Y \rightsquigarrow Y'$ such that

$$\begin{array}{ccc} X & \rightsquigarrow^q & Y \\ \downarrow a & & \downarrow b \\ X' & \rightsquigarrow^{q'} & Y' \end{array} \quad q' \circ a = b \circ q \quad \text{in } \text{Stoch}.$$

We call a and b the *legs* of the square. Squares compose componentwise, $(a', b') \circ (a, b) = (a' \circ a, b' \circ b)$, and $(\text{id}_X, \text{id}_Y)$ is a square from q to itself; since the deterministic morphisms form a subcategory $\text{Stoch}_{\text{det}}$ (§3), this defines a category $\text{Stoch}_{\text{det}}^{\rightarrow}$ whose objects are *all* the morphisms of Stoch and whose morphisms are the squares. Two subcategories are needed below:

- (i) $\text{Stoch}_{\text{det}, \text{cop}}^{\rightarrow}$, the full subcategory on the objects $q: X \rightsquigarrow Y$ whose domain X is isomorphic in Stoch to a copower $A \cdot I$ of the monoidal unit;
- (ii) $\text{Stoch}_{\delta}^{\rightarrow}$, the wide subcategory whose morphisms are the squares with both legs in the image of the Dirac embedding δ (Definition 3.2); it is closed under composition because δ is a functor.

Lemma 4.9 (Pointwise form of a square). *Let $f: X \rightarrow X'$ and $g: Y \rightarrow Y'$ be measurable maps. Then $(\delta f, \delta g)$ is a square from q to q' if and only if*

$$q'(f(x)) = g_* q(x) \quad \text{in } \mathcal{P}(Y'), \text{ for every } x \in X.$$

Proof. By (8), $(q' \circ \delta f)(x) = q'(f(x))$; by (7), $(\delta g \circ q)(x) = g_* q(x)$. $\square$

Remark 4.10 (Deterministic legs versus Dirac legs). $\text{Stoch}_{\delta}^{\rightarrow} \subseteq \text{Stoch}_{\text{det}}^{\rightarrow}$ because Dirac kernels are deterministic, and the inclusion is an equality in FinStoch . In general the deterministic morphisms into a space Z are all Dirac kernels exactly when every $\{0, 1\}$ -valued measure on Z is a Dirac measure, the converse recorded in Definition 3.2; if some $\{0, 1\}$ -valued measure ν on Z is not Dirac, the constant kernel at ν is deterministic but not in the image of δ . Hence every square with legs into X' and Y' lies in $\text{Stoch}_{\delta}^{\rightarrow}$ when this holds for X' and for Y' , in particular when they satisfy hypothesis (b) of Proposition 4.11. Hypothesis (b) is sufficient but not necessary: hypothesis (a) of the same proposition describes discrete spaces on which every $\{0, 1\}$ -valued measure is Dirac although the σ -algebra need not be countably generated. The parameter set of the one-way layout is a case in point: $\mathbb{R}^{\omega} \times \mathbb{R}_{>0}$ with the discrete σ -algebra, which is what $\Theta_0 \cdot I$ is for $\Theta_0 = U_K \Theta(\omega)$ of Example 2.5, has the cardinality of the continuum, below the first measurable cardinal, so every countably additive $\{0, 1\}$ -valued measure on it is Dirac; but a countably generated σ -algebra has at most continuum many elements, while the discrete one has $2^{\mathbb{R}}$.

Proposition 4.11 (Object-level form of the identification). *Transposition across (9) extends to a functor*

$$\Delta: \text{StatMod} \longrightarrow \text{Stoch}_{\text{det}}^{\rightarrow}, \quad (\Theta_0, S, p) \longmapsto (\widehat{p}: \Theta_0 \cdot I \rightsquigarrow S),$$

with the following properties.

1. Δ is injective on objects, and restricts to a bijection between the objects of StatMod and those arrows of Stoch whose domain is a copower $A \cdot I$ of the monoidal unit; consequently Δ is essentially surjective onto $\text{Stoch}_{\text{det, cop}}^{\rightarrow}$, whose objects are the arrows with domain merely isomorphic to such a copower (Definition 4.8).
2. Δ is bijective on hom-sets: for objects $x = (\Theta_0, S, p)$ and $x' = (\Theta'_0, S', p')$, the map $\text{StatMod}(x, x') \rightarrow \text{Stoch}_{\text{det}}^{\rightarrow}(\Delta x, \Delta x')$ induced by Δ is a bijection provided
 - (a) every countably additive $\{0, 1\}$ -valued measure on $(\Theta'_0, 2^{\Theta'_0})$ is a Dirac measure, automatic when $|\Theta'_0|$ is smaller than the first measurable cardinal (Lemma A.2), hence always in the finite and countable cases; and
 - (b) for a measurable space $Z: \Sigma_Z$ is countably generated and separates points, so that for every W the deterministic morphisms $W \rightsquigarrow Z$ are exactly the Dirac kernels of measurable maps [14, Ex. 10.4]. We refer to this as “(b) for Z ”; what is required here is (b) for S' .
3. Let $\text{StatMod}_{(a),(b)} \subseteq \text{StatMod}$ be the full subcategory on the objects (Θ_0, S, p) with Θ_0 satisfying (a) and S satisfying (b). Then Δ restricts to an equivalence between $\text{StatMod}_{(a),(b)}$ and the full subcategory of $\text{Stoch}_{\text{det, cop}}^{\rightarrow}$ on the arrows $X \rightsquigarrow S$ with S satisfying (b) and $X \cong A \cdot I$ for A satisfying (a). When all data are finite, write $\text{StatMod}_{\text{fin}}$ for the full subcategory on finite Θ_0 and finite discrete S , (a) and (b) are automatic, every object of FinStoch is a copower of I , and $\text{StatMod}_{\text{fin}} \simeq \text{FinStoch}_{\text{det}}^{\rightarrow}$ unconditionally.

Proof idea. With $A \cdot I := (A, 2^A)$, transposition (9) gives the object bijection of part (1); passing to $\text{Stoch}_{\text{det, cop}}^{\rightarrow}$ requires knowing that an isomorphism $A \cdot I \cong X$ of Stoch is deterministic, which is proved from the copower side, where singletons are measurable. On morphisms, $(r, f) \mapsto (\delta r, \delta f)$, the square commuting by Lemma 4.9; hypotheses (a) and (b) are exactly what is needed to run that lemma backwards, turning the two deterministic legs of a square into a function and a measurable map. The complete proof, together with the facts about deterministic morphisms and $\{0, 1\}$ -valued measures it relies on, is given in Appendix A. $\square$

The finite case is routine. The appeal to Ulam’s theorem in (a) should be carried as an explicit set-theoretic hypothesis, or avoided by restricting parameter objects to standard Borel spaces throughout, which incurs no statistical cost.

Remark 4.12 (Why the legs are deterministic). Definition 4.8 is not the most obvious choice. Write $\text{Stoch}^\rightarrow$ for the full arrow category, whose morphisms $q \rightarrow q'$ are *arbitrary* pairs of kernels (a, b) with $q' \circ a = b \circ q$; this is the definition one writes down first, and $\text{Stoch}_{\text{det}}^\rightarrow$ is a wide, non-full subcategory of it. Three points on why the restriction is imposed here.

(i) *It is an image condition, inherited from the domain.* The morphisms of StatMod are pairs of functions, and those of its measurable analogue $\text{StatMod}_{\text{Meas}}$, introduced in Remark 4.13(ii), pairs of measurable maps, and transposition carries both to pairs of Dirac kernels. The image of Δ therefore lies in $\text{Stoch}_\delta^\rightarrow$, which coincides with $\text{Stoch}_{\text{det}}^\rightarrow$ under the conditions recorded in Remark 4.10, hypothesis (b) of Proposition 4.11 among them; this is the precise sense in which the legs of Definition 4.8 are the transposed maps. Admitting arbitrary pairs would make Δ non-full for trivial reasons and leave part (2) of that proposition without content. More structurally: Brøns’ category is a comma category over a base equipped with a probability functor, and the legs of its squares are morphisms of that base; determinism is what survives of the base under transposition.

(ii) *Two mechanisms are lost.* The pointwise form of the commutation condition (Lemma 4.9) rests on (7) and (8), hence on the legs being Dirac; for stochastic legs there is no corresponding condition in Meas , and with it goes the fact that a diagram of deterministic morphisms commutes in Stoch if and only if the underlying diagram commutes in Meas (§3, fact (ii)), the mechanism by which $\text{Stoch}_{\text{det}}^\rightarrow$ is a faithful shadow of the Meas -level picture. Second, strict Kleisli equality, which is the right condition for deterministic legs, is not obviously the right one once the legs randomise: the ambient notion of equality in categorical probability is almost-sure equality relative to a state [14, §13], and adopting it would make $\text{Stoch}^\rightarrow$ a bicategory and any identification a biequivalence. Neither obstacle is fatal, and both lie outside what is needed here.

(iii) *The stochastic variant is a different subject, namely Le Cam’s.* Discussing [30], Huber recalls that handling sufficiency forced Le Cam to replace pointwise maps by randomised maps, and that the analysis of variance forced randomisation of the parameter spaces as well, so that morphisms of experiments became equivalence classes of pairs of transition functions [23]. His example separates the two categories. Let A be the experiment observing $X = (X_1, \dots, X_n)$, $n \geq 2$, i.i.d. $\mathcal{N}(\mu, \sigma^2)$, and B the experiment observing independent (U, V) with $U \sim \mathcal{N}(\mu, \sigma^2/n)$ and $V/\sigma^2 \sim \chi_{n-1}^2$: two models with parameter set $\mathbb{R} \times \mathbb{R}_{>0}$, on $\mathbb{R}^n$ and on $\mathbb{R} \times \mathbb{R}_{>0}$ respectively. The statistic $t = (\bar{X}, S^2)$, $S^2 = \sum_i (X_i - \bar{X})^2$, pushes each $P_A(\mu, \sigma^2)$ to $P_B(\mu, \sigma^2)$, so $(\text{id}, \delta t)$ is a morphism $A \rightarrow B$ in $\text{Stoch}_{\text{det}}^\rightarrow$. There is no morphism $B \rightarrow A$ in $\text{Stoch}_{\text{det}}^\rightarrow$, whatever its parameter leg: if $g: \mathbb{R} \times \mathbb{R}_{>0} \rightarrow \mathbb{R}^n$ were measurable with $g_* P_B(\theta) = P_A(a(\theta))$ for all θ , the direction $(g - \bar{g}(1, \dots, 1))/S_g$ of the sample $g(U, V)$ would be a statistic of B whose law, uniform on the unit sphere of the hyperplane $\sum_i x_i = 0$, is free of θ ; by Basu’s theorem [2] it would be independent of the complete sufficient statistic (U, V) , of which it is a function, hence almost surely constant, which it is not. In $\text{Stoch}^\rightarrow$, by contrast, the conditional distribution k of X given $(\bar{X}, S^2)$, free of the parameter, gives a morphism $(\text{id}, k): B \rightarrow A$, with $t \circ k = \text{id}_B$ but $k \circ t \neq \text{id}_A$: $k \circ t$ sends x to the uniform distribution on the sphere through x . The two become mutually inverse only after identifying morphisms that agree once composed with the model, $k \circ t \circ p_A = p_A$, Huber’s “suitably defined equivalence classes”. Allowing stochastic legs thus coarsens the isomorphism relation on objects to Blackwell equivalence: two models $p, p': \Theta \rightsquigarrow \cdot$ on the same parameter set are *Blackwell equivalent* when each is a garbling of the other, $p' = c \circ p$ and $p = c' \circ p'$ for kernels c, c' , Blackwell’s “equally informative” [7], deficiency zero in Le Cam’s theory [26], and mutual dominance for the informativeness preorder of [16, §5.1]. In the quotient just described, a morphism $B \rightarrow A$ over id_Θ with an inverse is exactly such a pair of garblings, so isomorphism there is Blackwell equivalence, and the invariant under study becomes informativeness rather than meaningfulness. That is the wrong invariant for the present purpose: Definition 5.1 asserts that reducing a design or reparametrising introduces no randomness, and a category whose reductions may themselves randomise cannot state it. The two readings of McCullagh’s question, which models are meaningful, and which experiments are equivalent, are genuinely different questions, and this paper is committed to the first.

Remark 4.13 (What this settles, and what it does not). Three points, in increasing order of consequence.

(i) *The essential image is not everything.* StatMod is *not* equivalent to $\text{Stoch}_{\text{det}}^{\rightarrow}$: the image is the full subcategory on arrows out of copowers of I . Any statement of the form “Brøns’ category is the category whose objects are the morphisms of Stoch ” is therefore false as stated and must carry the domain restriction. The restriction is invisible in the finite case only because there every object of FinStoch is a copower of I .

(ii) *The restriction is an artefact of the Set-valued formulation.* Replace Brøns’ $\text{StatMod} = (\text{id}_{\text{Set}} \downarrow \mathcal{P})$ by its measurable analogue $\text{StatMod}_{\text{Meas}} := (\text{id}_{\text{Meas}} \downarrow \mathcal{G})$, whose objects are measurable maps $\Theta \rightarrow \mathcal{G}S$. By (4) these are exactly the morphisms $\Theta \rightsquigarrow S$ of Stoch , with no copower condition, and the same argument gives a functor $\Delta_{\text{Meas}}: \text{StatMod}_{\text{Meas}} \rightarrow \text{Stoch}_{\text{det}}^{\rightarrow}$ (Definition 4.8, whose objects are arbitrary, not deterministic, morphisms of Stoch). The two levels behave differently and it is worth separating them.

On objects, the identification is a tautology, by (4): the copower restriction disappears because every measurable space, and not only a discrete one, is now admissible as a parameter object.

On morphisms it is not a tautology. Transposition sends a morphism (u, h) of $\text{StatMod}_{\text{Meas}}$, a pair of *measurable* maps, to the square $(\delta u, \delta h)$, and this assignment is unconditionally bijective on objects and full onto $\text{Stoch}_{\delta}^{\rightarrow}$. It is *faithful* exactly when δ is injective on the measurable maps concerned, that is when $\Sigma_{\Theta'}$ and $\Sigma_{S'}$ separate points; and $\text{Stoch}_{\delta}^{\rightarrow} = \text{Stoch}_{\text{det}}^{\rightarrow}$ exactly when every deterministic morphism into Θ' or into S' is a Dirac kernel (Remark 4.10), which is the remaining content of (b). Both requirements concern *both* legs, whereas Proposition 4.11 invokes (b) for S' alone, hypothesis (a) doing the parameter-side work there, at the price of the copower restriction. The correct statement is therefore: Δ_{Meas} restricts to an equivalence between the full subcategory of $\text{StatMod}_{\text{Meas}}$ on the objects whose parameter space and sample space both satisfy (b) and the full subcategory of $\text{Stoch}_{\text{det}}^{\rightarrow}$ on the arrows between such spaces; unconditionally one has only a bijective-on-objects full functor $\text{StatMod}_{\text{Meas}} \rightarrow \text{Stoch}_{\delta}^{\rightarrow}$.

Hypothesis (a) does not survive this passage, but neither does it drop out: it served to recognise a deterministic morphism out of a *discrete* parameter object as a Dirac measure, and in the measurable setting that task is taken over by (b) for Θ' . The two are in fact instances of a single requirement, the one isolated in Definition 3.2: that the $\{0, 1\}$ -valued measures on the codomain be exactly its points. Condition (b) secures it when the σ -algebra is countably generated and separating. Condition (a) is what remains of it over a discrete parameter object $(\Theta'_0, 2^{\Theta'_0})$, whose σ -algebra separates points but is not countably generated unless Θ'_0 is countable, so that the identification of $\{0, 1\}$ -valued measures with points degenerates into a set-theoretic question about measurable cardinals rather than a measure-theoretic one. Passing to $\text{StatMod}_{\text{Meas}}$ therefore does not remove a hypothesis: it replaces the set-theoretic instance by the measure-theoretic one, and is repaid by the disappearance of the copower restriction. No copower intervenes on either side. Nor does any strengthening of the hypotheses give an equivalence with the full arrow category $\text{Stoch}^{\rightarrow}$: the morphisms of a comma category over Meas are pairs of measurable maps, so their transposes are deterministic by construction, and the genuinely stochastic squares are never in the image (Remark 4.12).

So the unrestricted identification is correct *for the measurable comma category* on the full subcategory of objects satisfying (b) on both legs, and incorrect for the **Set**-valued one, where it acquires a domain restriction. The mathematical content of the identification is thus not the equivalence itself but the comparison functor $\text{StatMod} \rightarrow \text{StatMod}_{\text{Meas}}$, which is precisely the measurability gap of Remark 4.4, met from the other side. That functor is constructed in Proposition 4.14 below. It exists unconditionally, is fully faithful, and fails to be essentially surjective exactly at the non-discrete parameter objects; so the copower restriction of (i) is the non-invertibility of a counit, and it is invisible in the finite case for that reason.

(iii) *This is a statement about StatMod, not about models.* A McCullagh–Brøns model is a diagram $\widehat{P}: \mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \text{StatMod}$ (Proposition 2.7), and the parameter object at ω supplied by Theorem 4.1 is $\delta\Theta_{\text{Meas}}(\omega)$, a copower of I exactly when $\Theta_{\text{Meas}}(\omega)$ is discrete. The two identifications therefore agree in the finite case and diverge in general. Corollary 4.15 says how far the diagram-level statement can be pushed, and Proposition 4.11 should not be presented as going further than that.

The comparison functor of (ii) can be written down, and doing so settles more than the passage between the two formulations. Write $D: \mathbf{Set} \rightarrow \mathbf{Meas}$ for the functor equipping a set with its power-set

σ -algebra and $U: \mathbf{Meas} \rightarrow \mathbf{Set}$ for the underlying-set functor.

Proposition 4.14 (Discretisation of the parameter; $\mathbf{StatMod}$ as a coreflective subcategory of $\mathbf{StatMod}_{\mathbf{Meas}}$). $D \dashv U$, and $UG = \mathcal{P}$ by construction. Write $\varepsilon: DU \Rightarrow \text{id}_{\mathbf{Meas}}$ for the counit, the component ε_X is the identity function on UX , measurable because its domain is discrete, and put $\lambda := \varepsilon_{\mathcal{G}}: D\mathcal{P} \Rightarrow \mathcal{G}$. Then:

1. $\Phi: \mathbf{StatMod} \rightarrow \mathbf{StatMod}_{\mathbf{Meas}}, (\Theta_0, S, p) \mapsto (D\Theta_0, S, \lambda_S \circ Dp), (u, h) \mapsto (Du, h)$, is a functor; its structure map is p itself, read as a measurable map on a discrete domain.
2. $\Psi: \mathbf{StatMod}_{\mathbf{Meas}} \rightarrow \mathbf{StatMod}, (\Theta, S, q) \mapsto (U\Theta, S, Uq), (a, h) \mapsto (Ua, h)$, is a functor, and $\Phi \dashv \Psi$.
3. $\Psi\Phi = \text{id}_{\mathbf{StatMod}}$. Hence the unit is an identity, Φ is fully faithful, and $\mathbf{StatMod}$ is a coreflective subcategory of $\mathbf{StatMod}_{\mathbf{Meas}}$ with coreflector Ψ .
4. The counit of $\Phi \dashv \Psi$ at (Θ, S, q) is $(\varepsilon_\Theta, \text{id}_S)$, invertible if and only if $\Sigma_\Theta = 2^\Theta$. The essential image of Φ is the full subcategory of $\mathbf{StatMod}_{\mathbf{Meas}}$ on the objects whose parameter space is discrete.

None of this uses hypothesis (a) or (b) of Proposition 4.11, or any set-theoretic assumption.

Proof. $D \dashv U$ because every function out of a discrete space is measurable, so $\mathbf{Meas}(DA, X) = \mathbf{Set}(A, UX)$ naturally; $UG = \mathcal{P}$ because the underlying set of $\mathcal{G}S$ is $\mathcal{P}(S)$ and the underlying function of $\mathcal{G}(h)$ is pushforward. As a whiskering of ε , λ is natural, and each λ_S is the identity on underlying sets.

(1) $Dp: D\Theta_0 \rightarrow D\mathcal{P}(S)$ is measurable (both spaces discrete) and λ_S is measurable, so $\lambda_S \circ Dp$ is an object of $\mathbf{StatMod}_{\mathbf{Meas}}$. For a morphism (u, h) , using naturality of λ , functoriality of D , and the $\mathbf{StatMod}$ square $\mathcal{P}(h) \circ p = p' \circ u$,

$$\mathcal{G}(h) \circ \lambda_S \circ Dp = \lambda_{S'} \circ D(\mathcal{P}(h)) \circ Dp = \lambda_{S'} \circ D(p' \circ u) = (\lambda_{S'} \circ Dp') \circ Du,$$

which is the $\mathbf{StatMod}_{\mathbf{Meas}}$ square. Identities and composites are preserved componentwise.

(2) Ψ is well defined since $Uq: U\Theta \rightarrow U\mathcal{G}S = \mathcal{P}(S)$. For the adjunction, a morphism $\Phi(\Theta_0, S, p) \rightarrow (\Theta', T, q)$ is a pair (a, h) with $a: D\Theta_0 \rightarrow \Theta'$ measurable and $q \circ a = \mathcal{G}(h) \circ \lambda_S \circ Dp$. Transposing a across $D \dashv U$ gives $u = Ua: \Theta_0 \rightarrow U\Theta'$, and since both sides of the square are measurable maps out of the discrete space $D\Theta_0$ and U is faithful, the square is equivalent to its image under U , namely $Uq \circ u = \mathcal{P}(h) \circ p$, the $\mathbf{StatMod}$ square for $(u, h): (\Theta_0, S, p) \rightarrow \Psi(\Theta', T, q)$. The correspondence is bijective, and natural in both variables because on the parameter leg it is the $D \dashv U$ bijection and on the sample leg it is the identity.

(3) $UD = \text{id}_{\mathbf{Set}}$ and $U\lambda_S = \text{id}$, so $\Psi\Phi$ is the identity on objects and on morphisms. A left adjoint whose unit is invertible is fully faithful, and its domain is then coreflective in its codomain.

(4) $\Phi\Psi(\Theta, S, q) = (DU\Theta, S, \lambda_S \circ D(Uq))$, and $(\varepsilon_\Theta, \text{id}_S)$ is a morphism from it to (Θ, S, q) : both legs of the required square are measurable maps $DU\Theta \rightarrow \mathcal{G}S$ with underlying function Uq , and U is faithful. It is the counit by the triangle identities. Since ε_Θ is a bijection, it is an isomorphism in $\mathbf{Meas}$ exactly when Σ_Θ is the full power set; and $X \cong DA$ in $\mathbf{Meas}$ forces $\Sigma_X = 2^X$, which gives the description of the essential image. Fullness of that subcategory is automatic from (3). $\square$

The point of (4) is that the domain restriction which had to be carried through Proposition 4.11 by hand is a single counit, and one whose failure to be invertible is exactly non-discreteness of the parameter. Transporting to diagrams is then formal.

Corollary 4.15 (Diagram-level form). Let $\mathbf{C}$ be a small category. Post-composition with Φ and Ψ gives functors Φ_*, Ψ_* between the diagram categories $[\mathbf{C}, \mathbf{StatMod}]$ and $[\mathbf{C}, \mathbf{StatMod}_{\mathbf{Meas}}]$ with $\Phi_* \dashv \Psi_*$ and $\Psi_*\Phi_* = \text{id}$. Consequently, taking $\mathbf{C} = \mathbf{V} \times \mathbf{D}^{\text{op}}$:

1. Brøns' bijection (Proposition 2.7) transports along Φ_* to a bijection between McCullagh–Brøns models and those functors $\mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \mathbf{StatMod}_{\mathbf{Meas}}$ whose two projections are $D \circ U_K \Theta \text{pr}_\Omega^{\text{op}} \text{pr}_2$ and $\Gamma(\text{id} \times \text{pr}_\Omega^{\text{op}})$; the projection part of Proposition 2.7 is thus replaced by its discretisation on the parameter leg.

2. A diagram in $\text{StatMod}_{\text{Meas}}$ lies in the essential image of Φ_* if and only if it is pointwise discrete on the parameter side, the obstruction at each index being the counit component of Proposition 4.14(4).

In particular, the diagram-level identification holds up to a coreflection, and pointwise discreteness of the parameter object is not a technical convenience but exactly its boundary.

Proof. Post-composition with an adjunction is an adjunction between diagram categories, with unit and counit computed pointwise, and $\Psi_*\Phi_* = (\Psi\Phi)_* = \text{id}$. (1) A functor $F: \mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \text{StatMod}_{\text{Meas}}$ with the stated projections has parameter legs of the form $D\Theta_0$, so $F = \Phi_*\Psi_*F$ (both sides have the same underlying functions, and a map out of a discrete space is measurable); hence $F \mapsto \Psi_*F$ and $G \mapsto \Phi_*G$ are mutually inverse between such functors and the functors $\mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \text{StatMod}$ with projections $U_K \Theta \text{pr}_\Omega^{\text{op}} \text{pr}_2$ and $\Gamma(\text{id} \times \text{pr}_\Omega^{\text{op}})$, and Proposition 2.7 identifies the latter with models. (2) If F is pointwise discrete on the parameter side, each counit component $(\varepsilon_{\Theta_c}, \text{id})$ is invertible by Proposition 4.14(4), so $\Phi_*\Psi_*F \Rightarrow F$ is a natural isomorphism and F lies in the essential image; conversely Φ_*G is pointwise discrete by construction, and pointwise discreteness is invariant under isomorphism in $[\mathbf{C}, \text{StatMod}_{\text{Meas}}]$, since $X \cong DA$ in Meas forces $\Sigma_X = 2^X$. $\square$

Two consequences for how the open problem should now be stated. First, the object-level identification of Proposition 4.11 and the diagram-level statement are not competing formulations to choose between: they are the restriction and the transport of one adjunction, and the question of which is “correct” does not arise. Second, the passage from StatMod to $\text{Stoch}_{\text{det}}^{\rightarrow}$ factors as $\Delta \cong \Delta_{\text{Meas}}\Phi$, the isomorphism being the canonical identification $D\Theta_0 \cong \Theta_0 \cdot I$ in Stoch , and the two factors behave quite differently. The first, $\Phi: \text{StatMod} \rightarrow \text{StatMod}_{\text{Meas}}$, is unconditional. Every hypothesis in Proposition 4.11, the measurable-cardinal condition (a) and the countably-generated condition (b), is used only on the second, $\Delta_{\text{Meas}}: \text{StatMod}_{\text{Meas}} \rightarrow \text{Stoch}_{\text{det}}^{\rightarrow}$. What remains open is therefore confined to Δ_{Meas} .

Whether Brøns’ equivalence (Proposition 2.7), transported as in Corollary 4.15, continues to hold with $\text{Stoch}_{\text{det}}^{\rightarrow}$ in place of $\text{StatMod}_{\text{Meas}}$, and under which hypotheses Δ_{Meas} , hence the composite $\Delta \cong \Delta_{\text{Meas}}\Phi$ onto $\text{Stoch}_{\text{det, cop}}^{\rightarrow}$, is an equivalence, is left open here.

5 Meaningfulness as naturality

Two criteria are on the table, and it is worth keeping them apart before merging them. Tjur’s discussion [36] of [30] proposes an elementary one: a parameter function, a rule assigning to each finite sample of covariate values a parameter of the corresponding model, in his words a “function of the unknown distribution” [36, p. 1299], is *meaningful* when it is invariant under the formation of marginal models, that is, when reducing the design by removing some of the sampled x ’s leaves the associated parameter unchanged. The removal deletes sample points together with their covariate values, not points of a covariate space; Lemma 5.2 identifies its action on the quantities below with naturality over the insertions. McCullagh’s own criterion [30, §4.5] is that a subparameter be a natural transformation of functors on the design category.¹ Tjur quantifies over removal of sample points, acting, by Lemma 5.2, through the insertions alone; McCullagh over all design morphisms. Neither source asserts that the two coincide as mathematical requirements, and in general they do not, but how far they differ depends on a convention, and it is worth being explicit about whose. McCullagh takes the morphisms of Ω to be injective [30, §4.1], so under his own convention merging is not a morphism at all, and the gap between his criterion and Tjur’s consists exactly of the bijections—that is, of relabelling. Definition 2.1 follows Brøns in imposing no such restriction, and every map of covariate spaces factors as a surjection followed by an injection; so in the

¹The two indexings agree on the families compared here, so the comparison below is over a single class of morphisms. Designs index McCullagh’s components in §4.5 and natural over all design morphisms (ψ_d, ψ_c) , whereas Definition 5.1 indexes quantities by objects of Ω and quantifies over its morphisms; on families factoring through pr_Ω the squares with identity covariate leg are trivial, and the remaining ones are exactly the morphisms of Ω , since for any $\psi: \omega \rightarrow \omega'$ and any design (u, ω, x) the pair $(\text{id}_u, \psi): (u, \omega, x) \rightarrow (u, \omega', \psi \circ x)$ is a morphism of $\mathbf{D}$, so that pr_Ω is surjective on morphisms.

setting of this paper the three quantifier classes are nested strictly,

$$\underbrace{\text{insertions}}_{\text{Tjur}} \subsetneq \underbrace{\text{all injections (insertions and bijections)}}_{\text{McCullagh as stated}} \subsetneq \underbrace{\text{all of } \Omega}_{\text{naturality here}},$$

and the corresponding conditions are ordered the other way, naturality over all of Ω being the strongest. The outer step is merging; the inner step is relabelling. Here and throughout, the *insertions* are the proper inclusions $\omega' \subsetneq \omega$ between objects of Ω , the covariate legs of Tjur’s reductions, as fixed in Example 2.5, and $\Omega_{\text{red}} \subseteq \Omega$ denotes the subcategory they generate, whose morphisms are the inclusions (proper or identity). Insertions carry the surviving labels unchanged; an injection that is not an inclusion is a relabelling followed by an insertion, and naturality along it is not part of Tjur’s condition. What the bridge buys is that all three conditions become computations in the same ambient category so that the gap can be measured rather than elided, and, as §6 shows for the one-way layout, it is the reduction morphisms that carry the defect in the motivating example, so that Tjur’s weaker criterion already suffices to detect it there.

Throughout we take *meaningfulness* in the strong (naturality) sense, and record where the weaker Tjur condition would do. We fix the definition in the bridged setting.

The value space of a quantity may itself depend on the design, and a change of design then transports values: in the one-way layout (Example 2.5, ω a finite label set) the vector of group means at ω' is a point of $\mathbb{R}^{\omega'}$, carried along $\psi: \omega \rightarrow \omega'$ to $\mathbb{R}^{\omega}$ by the restriction ψ^* ; in the regression example (Example 2.8) the coefficient vector at ω' is a point of ω'^* , carried to ω^* by the transpose of ψ . The target of a design-indexed quantity is accordingly a functor $\Lambda: \Omega^{\text{op}} \rightarrow \text{Meas}$, of the same variance as Θ_{Meas} , so that the squares below compose. A constant Λ recovers the familiar fixed-scale case (Example 5.5 and the whole of §6); Example 5.6 has a genuinely varying target.

Definition 5.1 (Design-indexed quantity; meaningfulness). Let $\mathbf{P}: \mathbf{L} \Rightarrow \mathbf{R}$ be a bridged model (Theorem 4.3). Let $\Lambda: \Omega^{\text{op}} \rightarrow \text{Meas}$ be a functor, the *target*: $\Lambda(\omega)$ is the measurable space of values at design ω . A *design-indexed quantity* for $\mathbf{P}$ with target Λ is a family of morphisms in Stoch

$$g_{\omega}: \Theta_{\text{Meas}}(\omega) \rightsquigarrow \Lambda(\omega), \quad \omega \in \text{ob } \Omega$$

(the deterministic case is when g is an ordinary parameter functional). The quantity is *meaningful* if g is a natural transformation $\delta \Theta_{\text{Meas}} \Rightarrow \delta \Lambda$ (deterministic case) or $\Theta_{\text{Meas}} \Rightarrow \Lambda$ read in Stoch (kernel case): i.e. for every $\psi: \omega \rightarrow \omega'$ in Ω ,

$$g_{\omega} \circ \delta \Theta_{\text{Meas}}(\psi) = \delta \Lambda(\psi) \circ g_{\omega'} \quad \text{in } \text{Stoch}. \quad (10)$$

A design-indexed quantity g is *Tjur-natural* when (10) holds for every ψ of Ω_{red} , the subcategory of inclusions generated by the insertion morphisms as fixed at the opening of this section; Lemma 5.2 warrants the name: the formation of marginal models is removal of sample points, and on quantities indexed over Ω it acts through the insertions and through nothing else.

Tjur reductions. Fix design data in Set as in Example 2.5, with Ω a category of finite sets such as Ω_{lab} : $\mathbf{C} = \text{Set}$, both J ’s the evident underlying-set functors, and $\mathbf{U}$ a full subcategory of $\text{FinSet}_{\text{inj}}$ containing the underlying sets of the objects of Ω (so that every subsample inclusion between such sets is a morphism of $\mathbf{U}$). The regression instantiations of §2.2, whose covariate objects are vector spaces, fall outside this setting. A *Tjur reduction* of a design (u, ω, x) with x surjective is a morphism of $\mathbf{D}$

$$(\varphi, \psi): (u', \omega', x') \longrightarrow (u, \omega, x)$$

in which $\varphi: u' \hookrightarrow u$ is a proper subsample inclusion, $\omega' = x(\varphi(u'))$ the surviving support, x' the corestriction of $x \circ \varphi$, and $\psi: \omega' \hookrightarrow \omega$ the inclusion: the removal of the sample points $u \setminus \varphi(u')$ together with the covariate values no longer observed, Tjur’s “removal of some of the x ’s” [36, p. 1299]. His condition at such a reduction compares the parameter assigned to the marginal model with the parameter for the bigger sample, the former being for him a function of the marginal distribution. Since a bridged model is natural at (φ, ψ) , this is square (2), the marginal law at the reduced design depends on θ only through $\Theta_{\text{Meas}}(\psi)\theta$; the comparison is therefore square (10) at ψ , and we say g *satisfies Tjur’s condition* at the reduction when that square commutes.

Lemma 5.2 (Tjur reductions act through their covariate shadows). *In the setting above, a design-indexed quantity g satisfies Tjur’s condition at every Tjur reduction of every design with a surjective design map if and only if g is Tjur-natural.*

Proof. If g is Tjur-natural: the covariate leg ψ of a Tjur reduction is the inclusion of the surviving support, hence an insertion (a proper inclusion) when the removal empties some fibre of x and the identity otherwise; square (10) holds at the former by hypothesis and at the latter trivially. Conversely, it suffices to check (10) at each insertion $\psi: \omega' \subseteq \omega$, since such squares paste along composites and hold trivially at identities. Consider the tautological design $(\omega, \omega, \text{id}_\omega)$, one unit per treatment label, with surjective design map, and the subsample $\varphi = \psi: \omega' \hookrightarrow \omega$, a morphism of $\mathbf{U}$ because ω' and ω are objects of Ω , hence of $\mathbf{U}$, and $\mathbf{U}$ is full in $\text{FinSet}_{\text{inj}}$. The surviving support is $x(\varphi(\omega')) = \omega'$, so $(\psi, \psi): (\omega', \omega', \text{id}) \rightarrow (\omega, \omega, \text{id})$ is a Tjur reduction with covariate leg ψ , and Tjur’s condition at it is square (10) at ψ . $\square$

Remark 5.3 (What the indexing screens off). We plainly state two limitations of the Lemma. First, the same removal recorded without corestriction, as $(\varphi, \text{id}_\omega): (u', \omega, x \circ \varphi) \rightarrow (u, \omega, x)$, has identity covariate leg, and for quantities indexed over Ω every such square is trivial (footnote 1); the force of Tjur’s criterion is carried entirely by the corestriction, that is, by his own requirement that the parameter of the marginal model be a function of the marginal distribution, which, for the models of §2.2, the surviving support parametrises. Second, Tjur’s parameter functions are indexed by samples, and before his criterion is imposed they include the sample size n and the average $\bar{x}$ weighted by multiplicities [36, p. 1299]; quantities in the sense of Definition 5.1 are indexed by Ω and can express neither, replication-dependence is screened off by the indexing rather than eliminated by the criterion, and its model-side trace is the $\mathbf{U}$ -slot failure of Example 2.16. What the criterion adjudicates on Ω -indexed quantities is dependence on the observed covariate values, and there, by the Lemma, sample-level invariance and naturality over Ω_{red} coincide.

Proposition 5.4 (Meaningfulness is naturality; comparison with Tjur’s criterion; type-level reading). *Let g be a design-indexed quantity. Then:*

1. *if g is deterministic in the sense $g_\omega = \delta f_\omega$ for measurable $f_\omega: \Theta_{\text{Meas}}(\omega) \rightarrow \Lambda(\omega)$ (an ordinary parameter functional), and the σ -algebras of the $\Lambda(\omega)$ separate points, then g is meaningful iff the f_ω form a natural transformation $\Theta_{\text{Meas}} \Rightarrow \Lambda$ in Meas ;*
- (1') *meaningfulness implies Tjur-naturality; the converse holds when $\Omega_{\text{red}} = \Omega$, and fails in general (Example 5.6);*
2. *meaningfulness is stable under postcomposition with natural transformations of targets $\Lambda \Rightarrow \Lambda'$ (in Meas , or in Stoch for kernel quantities) and under pullback along functors $F: \Omega' \rightarrow \Omega$ (change of covariate scheme; the bridged model $\mathbf{P}$ is restricted along F as well);*
3. *in the functor category $\text{Stoch}^{\Omega^{\text{op}}}$ (objects: functors; morphisms: natural transformations), g is meaningful iff it is a morphism $\delta\Theta_{\text{Meas}} \rightarrow \delta\Lambda$ (deterministic case) or $\Theta_{\text{Meas}} \rightarrow \Lambda$ (kernel case). A non-natural family is not a morphism of the category, it lacks the asserted type.*

Proof. (1) With $g = \delta f$, square (10) reads $\delta(f_\omega \circ \Theta_{\text{Meas}}(\psi)) = \delta(\Lambda(\psi) \circ f_{\omega'})$, and δ is faithful on maps into the point-separating space $\Lambda(\omega)$ (Definition 3.2), so this holds iff $f_\omega \circ \Theta_{\text{Meas}}(\psi) = \Lambda(\psi) \circ f_{\omega'}$. (1'): the implication and the equality case are immediate, Tjur-naturality being (10) with the quantifier restricted to Ω_{red} ; Example 5.6 supplies a quantity that is Tjur-natural but not meaningful (Example 5.5, by contrast, fails already on Ω_{red} and does not separate the two conditions). (2) is functoriality of pasting. (3) is a restatement of the definition of the functor category. $\square$

Example 5.5 (Tjur’s non-example: the design-averaged expectation). Recast in the present notation, Tjur’s own non-example [36, pp. 1299–1300] is the simple regression model with $\Theta(\omega) = \mathbb{R}^2 \times \mathbb{R}_{>0}$, coordinates $(\alpha, \beta, \sigma^2)$, Ω the category of finite covariate selections, and Λ constant at $\mathbb{R}$ (with $\Lambda(\psi) = \text{id}$);

this recasts the unreplicated case, the design map injective, so that sample points and covariate values coincide, the general case, with $\bar{x}$ weighted by multiplicities, is screened off by the Ω -indexing (Remark 5.3). Put

$$g_\omega(\alpha, \beta, \sigma^2) = \alpha + \beta \bar{x}_\omega, \quad \bar{x}_\omega = |\omega|^{-1} \sum_{x \in \omega} x.$$

Each g_ω is a legitimate parameter of the model at ω . But for an insertion $\psi: \omega \subseteq \omega'$ the parameter map $\Theta(\psi)$ is the identity, while $\bar{x}_\omega \neq \bar{x}_{\omega'}$ in general, so (10) reads $\alpha + \beta \bar{x}_\omega = \alpha + \beta \bar{x}_{\omega'}$ and fails off the locus $\beta = 0$. The failure is already visible on Ω_{red} , so the quantity is non-meaningful in both Tjur's sense and ours. The same argument, with $\Theta(\psi)$ again the identity, disqualifies the correlation coefficient, whose design dependence enters through the covariate variance [30, §6.2]; here it is the design-*scale* rather than the design-*mean* that fails to transport. In the bridged setting both failures are equalities that fail, not methodological unease.

Example 5.6 (A quantity natural on Ω_{red} but not on Ω). Let $\Omega = \Omega_{\text{lab}}$ (Example 2.5), let Θ be any parameter functor, and take as target $\Lambda(\omega) = \mathbb{R}^{\omega \times \omega}$ with $\Lambda(\psi) = (\psi \times \psi)^*$, i.e. $\Lambda(\psi)(v)_{(a,b)} = v_{(\psi a, \psi b)}$. Put $g_\omega(\theta)_{(a,b)} = [a \neq b]$, the indicator that two treatment labels are distinct. Square (10) at $\psi: \omega \rightarrow \omega'$ reads $[a \neq b] = [\psi a \neq \psi b]$ for all $a, b \in \omega$, which holds iff ψ is injective. Hence g is natural along every injection, in particular along every insertion, hence Tjur-natural, and natural even under McCullagh's criterion as stated (all injections, i.e. insertions and bijections), and fails along every merge, for instance along $p: \{1, 2\} \rightarrow \{*\}$: the gap between Tjur-naturality and meaningfulness is realised at the merges.

Corollary 5.7 (Finite verification). *Suppose Ω has finitely many morphisms, $\Theta_{\text{Meas}}(\omega)$ and $\Lambda(\omega)$ are finite (discrete) for every ω , and the entries of the kernels g_ω lie in a subfield $k \subseteq \mathbb{R}$ with decidable equality (for instance $\mathbb{Q}$, or the real algebraic numbers). Then meaningfulness of g is decidable: it is equivalent to the finite conjunction, over the morphisms $\psi: \omega \rightarrow \omega'$ of any generating set of Ω , of the matrix equalities (10), each between two $|\Theta_{\text{Meas}}(\omega')| \times |\Lambda(\omega)|$ stochastic matrices with entries in k .*

Proof. If (10) holds at ψ and at ψ' then it holds at $\psi'\psi$, by pasting the two squares ($\delta\Theta_{\text{Meas}}$ and $\delta\Lambda$ being functors), so it suffices to check a generating set, which is finite; each check is an equality of finitely many elements of k , decidable by hypothesis. $\square$

6 A cohomological formulation of coherence for finite designs

This section formulates, it does not yet fully prove, the connection between meaningfulness and the cohomology of diagrams of algebras, which links the present framework to the diagrammatic Hochschild/Gerstenhaber–Schack theory of [18, 4], as revisited in [10, §3]. The logical role of this section is precisely delimited: cohomology here *controls obstructions* to coherence; it is upstream of the bridge, not a proof of it. A distinct cohomological tradition attaches invariants to statistical structures directly; its relation to the complex constructed here is set out in Remark 6.10 at the end of the section.

6.1 The diagram of algebras of a finite design scheme

Let Ω be a finite category of covariate schemes and let all spaces, the parameter sets $\Theta(\omega)$ and the target sets $\Lambda(\omega)$, be finite, as in Corollary 5.7. Fix a field $k \supseteq \mathbb{Q}$.

Definition 6.1. Write Alg_k for the category of associative unital k -algebras and unital k -algebra homomorphisms. The *parameter diagram of algebras* is the functor

$$A: \Omega \longrightarrow \text{Alg}_k, \quad A(\omega) := k^{\Theta(\omega)}$$

(functions on the parameter set, a commutative k -algebra under pointwise operations), covariant because Θ is contravariant and $k^{(-)}$ is contravariant again: $A(\psi) = (\Theta(\psi))^*$ is precomposition. Similarly, each target functor Λ yields $B(\omega) = k^{\Lambda(\omega)}$.

A *deterministic* design-indexed quantity $g = \{g_\omega: \Theta(\omega) \rightarrow \Lambda(\omega)\}$ dualises to a family of algebra maps $g^* = \{g_\omega^*: B(\omega) \rightarrow A(\omega)\}$, and g is meaningful (Definition 5.1) iff g^* is a natural transformation of diagrams $B \Rightarrow A$: all spaces being finite and discrete, meaningfulness is naturality of the underlying maps (Proposition 5.4(1)), and $k^{(-)}$ is faithful, a map f is recovered from f^* through the indicator functions of its fibres, so naturality of g and of g^* are equivalent. Failure of naturality is measured, for each $\psi: \omega \rightarrow \omega'$, by the difference

$$c(\psi) := A(\psi)g_\omega^* - g_{\omega'}^*B(\psi) \in \text{Hom}_k(B(\omega), A(\omega')), \quad (11)$$

a 1-cochain on Ω in the sense of Baues–Wirsching [4, (1.4)], with coefficients in the natural system $D: F\Omega \rightarrow \text{Ab}$ obtained by pulling back the bimodule $M(\omega, \omega') = \text{Hom}_k(B(\omega), A(\omega'))$ along the forgetful functor $F\Omega \rightarrow \Omega^{\text{op}} \times \Omega$ [4, (1.16)–(1.18)]: $D(\psi) = \text{Hom}_k(B(\omega), A(\omega'))$ for $\psi: \omega \rightarrow \omega'$, a morphism $(\alpha, \beta): \psi \rightarrow \alpha\psi\beta$ of the factorization category acting by $f \mapsto A(\alpha)fB(\beta)$. With bimodule coefficients this is Hochschild–Mitchell cohomology [4, (8.5)]; when $B = A$ this complex is the Hochschild-degree-1 row of the Gerstenhaber–Schack bicomplex of the diagram A [10, Prop. 3.8] (defined over a poset in [18], with diagrams over a small category as in [19]). The group H^1 computed below is the cohomology of that row for its horizontal, Baues–Wirsching, differential, i.e. it lives in bidegree (1, 1) of the bicomplex; it is not Gerstenhaber–Schack cohomology in total degree 2, and no vertical (Hochschild) differential is used here. The relation of the bicomplex to Baues–Wirsching cohomology is controlled by a spectral sequence attributed to Robinson [33] in [10, §1, Thm. 3.11].

Remark 6.2 (Why H^1 is the right home). The cohomological formulation is forced rather than chosen. Meaningfulness of a design-indexed quantity is the assertion that the family g^* is an element of $\text{Nat}(B, A) = \int_\omega \text{Hom}_k(B(\omega), A(\omega))$, an end (the subgroup of $\prod_\omega \text{Hom}_k(B(\omega), A(\omega))$ cut out by the equations $A(\psi)h_\omega = h_{\omega'}B(\psi)$, one for each ψ ; [27, IX.5]); and H^0 of the Baues–Wirsching complex of the diagram is that end. The defect cochain (11) is then a 1-cocycle (Lemma 6.3) whose class in the first derived functor of that end [4, (8.5)], H^1 , is the obstruction to correcting g^* by an arbitrary 0-cochain, so H^1 is the only place where a non-meaningful family could leave a trace, and the class-relativity of Proposition 6.4(2) below is the statement that the 0-cochain group is too large for that trace to survive unless the admissible correction class is cut down.

Lemma 6.3 (Cocycle property). *The cochain c of (11) satisfies the 1-cocycle identity $c(\psi'\psi) = A(\psi')c(\psi) + c(\psi')B(\psi)$ for composable ψ, ψ' , and $c = 0$ iff g is meaningful.*

Proof. Expand $c(\psi'\psi)$ using functoriality of A, B and add-subtract the mixed term $A(\psi')g_{\omega'}^*B(\psi)$:

$$A(\psi'\psi)g_\omega^* - g_{\omega'}^*B(\psi'\psi) = A(\psi')(A(\psi)g_\omega^* - g_{\omega'}^*B(\psi)) + (A(\psi')g_{\omega'}^* - g_{\omega'}^*B(\psi'))B(\psi).$$

The second statement is the equivalence recorded after Definition 6.1: $c = 0$ says that g^* is natural, which holds iff g is. $\square$

Proposition 6.4 (Obstruction formulation). *Let g be a design-indexed quantity given only on the objects of Ω (a “raw family”), fix an admissible class $C \subseteq C^0 = \prod_\omega \text{Hom}_k(B(\omega), A(\omega))$ of 0-cochains, and suppose one seeks a correction $h \in C$ such that the corrected family $g^* + h$ is natural. (A 0-cochain is merely a family of linear maps; $g^* + h$ is the dual of a design-indexed quantity only when it is a family of algebra homomorphisms, which C must ensure, the recalibration class of Example 6.6 does so by construction.) Then:*

1. *a corrected natural family exists iff the equation $c = -d^0h$ is solvable with $h \in C$; for a linear subspace C this is finite linear algebra, and for a finite non-linear class (such as the recalibrations of Example 6.6) it is finite combinatorics. Both c and the correcting h , when it exists, are computed exactly;*
2. *the obstruction is intrinsically class-relative. When corrections range over the whole 0-cochain group, the obstruction vanishes identically and carries no information: by (11) the defect cochain of any raw family is by construction the coboundary d^0g^* of that family, so $[c_g] = 0$ in H^1 for every g whatever, with no hypothesis. Admissible classes must therefore be proper subsets of the 0-cochains to carry any obstruction at all;*

3. $H^1(\Omega; \text{Hom}(B, A))$ is accordingly not an obstruction group for individual quantities but an invariant of the design scheme (and of the target Λ), computable by finite linear algebra; Example 6.8 records a nonzero instance.

Remark 6.5 (Why part (2) matters). The identity $c_g = d^0 g^*$ is the reason the naive reading of part (1), “non-meaningfulness is certified by a nonzero class in H^1 ”, is false as stated, and the phrase “complete obstruction in H^1 ” should not be used without naming the class. Equivalently: with corrections ranging over all 0-cochains, the zero family is natural and corrects anything. The cocycle identity of Lemma 6.3 is then the identity $d^1 d^0 = 0$.

Proof. If $g^* + h$ is natural for some 0-cochain h then $c = -dh$, where d is the Baues–Wirsching differential $(dh)(\psi) = A(\psi)h_\omega - h_{\omega'} B(\psi)$, so $[c] = 0$ in H^1 , a statement carrying no information about C ; conversely if $c = -dh$ with $h \in C$ then $g^* + h$ is natural by reversing the computation. Part (2) is the substitution $h = -g^*$ in this equivalence, and part (3) follows since the complex is finite-dimensional in each degree. $\square$

The linear-algebraic skeleton (Lemma 6.3 and the displayed equivalence) is elementary, and the proof above establishes all three parts. The statement is discharged, in the class-relative form that part (2) forces, by an exact computation for the one-way layout; the findings are recorded in Example 6.6 and its successors.

6.2 The one-way layout as test case

Example 6.6 (One-way layout: the full finite complex). Truncate Example 2.5 to the smallest nontrivial covariate category. *The choice of that category is not a matter of convenience: the merge morphism alone cannot host the test case.* Write Ω_{arr} for the two-object category with $\omega_2 = \{1, 2\}$ (two treatment groups), $\omega_1 = \{*\}$ (groups merged) and the single non-identity morphism $p: \omega_2 \rightarrow \omega_1$, and Ω_{full} for the full subcategory of FinSet on the same two objects, which adds the injections $i_1, i_2: \omega_1 \rightarrow \omega_2$ (subgroup selection), the swap σ (treatment relabelling) and the constants $c_k = i_k p$: eight morphisms in all, with $p i_k = \text{id}$ and $\sigma i_1 = i_2$. Identifying $\omega_1 = \{*\}$ with $\{1\}$, i_1 is the insertion $\{1\} \subseteq \{1, 2\}$, the shadow of Tjur’s reduction to the first group, Lemma 5.2, and $i_2 = \sigma i_1$ is that insertion followed by a relabelling; the reduction to the second group is the insertion $\{2\} \subseteq \{1, 2\}$, which i_2 represents up to the same identification. Discretise the parameters: $\Theta(\omega) = M^\omega \times S$ with M, S finite sets of admissible means and dispersions, $\Theta(\psi)(\mu, s) = ((\mu, \mu), s)$ the diagonal. Then:

$$A(\omega_1) = k^{M \times S}, \quad A(\omega_2) = k^{M^2 \times S}, \quad A(\psi) = \text{restriction to the diagonal}.$$

The Baues–Wirsching 1-cochains with coefficients as above are indexed by the single non-identity morphism $\psi = p$ (identities carry no information after normalisation), so

$$C^0 = \text{Hom}_k(B(\omega_1), A(\omega_1)) \oplus \text{Hom}_k(B(\omega_2), A(\omega_2)), \quad C_{\text{norm}}^1 = \text{Hom}_k(B(\omega_2), A(\omega_1)),$$

the latter being the coefficient of the natural system at $\psi = p$, and H^1 is the cokernel of the explicit linear map $d^0(h_1, h_2) = A(\psi)h_2 - h_1 B(\psi)$. Everything is therefore a finite matrix computation once the candidate g is fixed. Take for g the *marginal* (total) dispersion under balanced allocation,

$$g_{\omega_1}(\mu, s) = s, \quad g_{\omega_2}((\mu_1, \mu_2), s) = s + \frac{(\mu_1 - \mu_2)^2}{4},$$

the one-way analogue of Tjur’s $\alpha + \beta \bar{x}$ of Example 5.5: a genuine parameter contaminated by a term that depends on which groups the design includes. The quantity is ours: it should not be confused with the overdispersion passage of Tjur’s discussion, which concerns whether a generalised linear model admits a scale extension leaving the link-linear estimates unchanged, a non-existence claim about a family of models, not a failure of naturality, and one on which nothing here bears. Throughout, the target functor Λ is constant at the finite set $T := g_{\omega_2}(\Theta(\omega_2)) \subset \mathbb{Q}$ of values of g (which contains $S = g_{\omega_1}(\Theta(\omega_1))$), with $\Lambda(\psi) = \text{id}_T$; accordingly $B(\omega) = k^T$ for both objects and $B(\psi) = \text{id}$. The computation was carried out in exact arithmetic, the defect cochain, the cocycle identity, and the corrected linear system over $\mathbb{Q}$; the two cochain-complex ranks modulo three 31-bit primes, in agreement, then re-derived exactly over $\mathbb{Q}$, and returns three findings.

Example 6.7 (Findings). (i) *The arrow category is blind.* Over Ω_{arr} the family above is *natural*: its defect cochain vanishes identically, because the between-group term $(\mu_1 - \mu_2)^2/4$ vanishes on the diagonal sub-model $\Theta(p)$. Moreover $H^1(\Omega_{\text{arr}}; \text{Hom}(B, A)) = 0$ (for $|M| = |S| = 2$: $\dim C^1 = \dim Z^1 = \dim B^1 = 16$), so no quantity whatever carries a class there. It is the subgroup-selection morphisms i_1, i_2 , not the merge, that detect the failure, which is to say that the failure is already visible to Tjur’s own criterion, quantified over removal of sample points alone: the defect at the insertion i_1 is a violation of his condition at the reduction to the first group (Lemma 5.2).

(ii) *Non-meaningfulness certified; a deterministic repair recovered.* Over Ω_{full} with $M = \{0, 1\}$, $S = \{1, 2\}$, so $T = \{1, \frac{5}{4}, 2, \frac{9}{4}\}$: the defect cochain is nonzero exactly on $\{i_1, i_2, c_1, c_2\}$ and zero on p and σ , and the cocycle identity of Lemma 6.3 was checked entrywise on all 22 composable pairs. Solving $c = -d^0 h$ over the full 0-cochain group, which by Proposition 6.4(2) must succeed, returns a corrected family that is again deterministic; the solution obtained by setting the free parameters of the row-reduced system to zero is the *within-group* dispersion $((\mu_1, \mu_2), s) \mapsto s$, i.e. the between-group term is discarded. The solution set is the affine space $-g^* + \text{Nat}(B, A)$ and contains other deterministic members, so no canonicity is claimed for this one; the vanishing of the unrestricted class (part (2) of the Proposition) is thus not a defect of the theory, and the statistically sensible repair is among the solutions.

(iii) *Nonzero obstruction relative to an admissible class.* Take as admissible the *information-bearing recalibrations*: corrected quantities $r_\omega \circ g_\omega$ with $r_\omega: T \rightarrow T$ and r_{ω_2} non-constant, so that the quantity may be re-expressed on its value scale but must remain a non-trivial function of the observed marginal dispersion. Naturality along i_1, i_2 reads $r_{\omega_1}(s) = r_{\omega_2}(s + (\mu_1 - \mu_2)^2/4)$ for all μ, s , and forces r_{ω_2} to be constant on the partition of T generated by $s \sim s + (\mu_1 - \mu_2)^2/4$, reducing feasibility to an exact union–find computation; the conditions along the other morphisms add nothing (naturality along σ holds for every recalibration since g_{ω_2} is symmetric; along p it reads $r_{\omega_2} = r_{\omega_1}$ on S , the case $\mu_1 = \mu_2$ of the condition along i_k ; and along $c_k = i_k p$ it follows from the latter). Call the design *scale-confounded* when some marginal value $s + (\mu_1 - \mu_2)^2/4$ with $\mu_1 \neq \mu_2$ is itself an admissible within-group dispersion, i.e. lies in S . With $M = \{0, 1\}$ and $S = \{1, \frac{5}{4}\}$, or already with $S = \{1, 2\}$ at three mean levels $M = \{0, 1, 2\}$, the partition collapses to a single block, every natural recalibration is constant, and the obstruction relative to this class is nonzero: no information-bearing recalibration of the marginal dispersion is natural. (With $M = \{0, 1\}$, $S = \{1, 2\}$ as in (ii) the partition has the two blocks $\{1, \frac{5}{4}\}$ and $\{2, \frac{9}{4}\}$, and a non-constant natural recalibration exists.)

Example 6.8 (A new invariant). Computing $H^1(\Omega; \text{Hom}(B, A))$ for its own sake, part (3) of Proposition 6.4, gives 0 for Ω_{arr} , and for Ω_{full} a group that vanishes at two levels of the treatment mean but not at three: with $M = \{0, 1, 2\}$, $S = \{1, 2\}$ and Λ constant at T as above (so $|T| = 5$), $\dim_k H^1(\Omega_{\text{full}}; \text{Hom}(B, A)) = 10 = 2|T|$, computed as in Example 6.6. The group depends on Λ as well as on Ω : with B constant and $B(\psi) = \text{id}$ it is the $|T|$ -fold direct sum of the cohomology with coefficients $\psi \mapsto A(\text{cod } \psi)$. This is the instance behind the phrase “nontrivial” in that part. Its statistical meaning is not known to us and is not pursued here.

Remark 6.9 (Scope of the cohomological claim). The correct summary of this section is: *for finite design schemes, meaningfulness of a design-indexed quantity is decidable, and its correctability within a prescribed statistically admissible class is controlled by a computable obstruction relative to that class.* The class-relativity is not a hedge but a theorem (Proposition 6.4(2)). This must not be conflated with a cohomological proof of the bridge: the bridge (Section 4) is proved by Kleisli transposition and needs no cohomology; the cohomology enters one level up, where coherence of quantities over designs is at stake.

Remark 6.10 (Information cohomology). A separate cohomological tradition attaches invariants to statistical structures directly: entropy is characterised as the essentially unique 1-cocycle of information cohomology in the finite discrete case [3] (for the continuous extension see [38]), as the unique (up to a non-negative scalar) functorial, convex-linear and continuous information-loss measure [1], as an operad derivation [8], and defined, in divergence-enriched Markov categories, as the divergence of a morphism from determinism [32]; for the functional equation underlying the cohomological characterisation see [5], and for cohomological models of information networks in the same tradition see [28]. The complex of

Definition 6.1 and the information-cohomological complexes have different coefficients and different indexing categories, and *no comparison map between them is constructed here*.

A prior question sits underneath any such comparison, and we state it separately because it is not cohomological. Information cohomology carries its own account of conditioning, organised around Shannon’s chain rule [3, 38], and touches statistical independence only through the vanishing of mutual information [3]; that account is not the one native to Markov categories, of which Section 3 recalls only the fragment used here: there, conditioning is disintegration in the sense of [11] (Bayesian inversion being the special case with a prior) and independence is the diagrammatic condition of [14]. Whether the two agree, and under what hypotheses, is to our knowledge not known; no translation between them is constructed or assumed anywhere in this paper, and none of its results depends on one existing. We record this because the comparison of the two formalisms at the level of these primitives is logically prior to any comparison of their cohomologies.

7 Scope and non-claims

For the record, two statements adjacent to this paper’s results are formulable in the present framework but are *not* established here or, to our knowledge, anywhere.

Remark 7.1 (Paradigms as morphism-shapes). Fisherian, Bayesian, nonparametric, statistical-learning and hybrid latent-variable modelling can each be *presented* as a choice of morphism-shape on the shared parameter–data structure of Section 4 (e.g. a latent-variable model as a composite through an unobserved object). These presentations organise the paradigms; they do not prove any equivalence between paradigm-specific theorems, and no such claim is made.

Remark 7.2 (Nonparametric and infinite-dimensional limits). All theorems above are stated for the parametric, measurably-regular case. Presenting nonparametric models as (co)limits of finite-dimensional ones in Stoch is a programme, not a technique available off the shelf: Stoch is not complete or cocomplete in the ways naive arguments require, and nothing here should be described as securing the nonparametric case.

A Proof of Proposition 4.11

Throughout, for a set A we write $A \cdot I$ for the measurable space $(A, 2^A)$. This is the coproduct in Meas of A copies of the one-point space, and δ preserves coproducts (Remark 4.7), so it is the copower of the monoidal unit in Stoch; with this convention, “the domain of q is a copower” means that the domain of q carries its full power set. For $p: \Theta_0 \rightarrow \mathcal{P}(S)$ the transpose $\widehat{p}: \Theta_0 \cdot I \rightsquigarrow S$ is the kernel $\widehat{p}(\theta)(B) = p(\theta)(B)$, measurable in θ because the domain is discrete; conversely every kernel $q: A \cdot I \rightsquigarrow S$ is $\widehat{q}$ for the function $\check{q}: a \mapsto q(a)$. This is (9), and the two assignments are mutually inverse.

The proof uses four facts about deterministic morphisms and one about $\{0, 1\}$ -valued measures on power sets. The first is recorded in §3 with a reference; the short proofs are included to keep the appendix self-contained.

Lemma A.1. *Let $k: X \rightsquigarrow Y$ be a morphism of Stoch.*

1. *k is deterministic if and only if $k(x)(B) \in \{0, 1\}$ for every $x \in X$ and every $B \in \Sigma_Y$ [14, Ex. 10.4].*
2. *If Σ_Y separates points, then δ is injective on $\text{Meas}(X, Y)$.*
3. *If Σ_Y is countably generated and separates points, then every $\{0, 1\}$ -valued probability measure on Y is δ_y for a unique $y \in Y$; consequently every deterministic $k: X \rightsquigarrow Y$ is δf for a unique measurable $f: X \rightarrow Y$.*
4. *If $v: A \rightsquigarrow X$ and $u: X \rightsquigarrow A$ are mutually inverse and v is deterministic, then so is u [14, Lem. 10.9(c)].*

Proof. (1) In Stoch the copy map is the Dirac kernel of the diagonal, so by (7) and (8) the kernel $\text{copy}_Y \circ k$ sends x to the image of $k(x)$ under the diagonal $Y \rightarrow Y \times Y$, while $(k \otimes k) \circ \text{copy}_X$ sends x to the product measure $k(x) \otimes k(x)$. On a measurable rectangle $B \times C$ these two measures take the values $k(x)(B \cap C)$ and $k(x)(B)k(x)(C)$ respectively, and since rectangles form a π -system generating $\Sigma_Y \otimes \Sigma_Y$, two probability measures on $Y \times Y$ agreeing on rectangles agree. Hence k is deterministic iff $k(x)(B \cap C) = k(x)(B)k(x)(C)$ for all x, B, C . Taking $B = C$ shows that each $k(x)$ is $\{0, 1\}$ -valued. Conversely, if μ is a $\{0, 1\}$ -valued probability measure then $\mu(B \cap C) = 1$ iff $\mu(B) = \mu(C) = 1$, if both are 1 then $\mu(Y \setminus (B \cap C)) \leq \mu(Y \setminus B) + \mu(Y \setminus C) = 0$, and if one is 0 then $\mu(B \cap C) = 0$, that is, $\mu(B \cap C) = \mu(B)\mu(C)$.

(2) $\delta f = \delta g$ means $\delta_{f(x)} = \delta_{g(x)}$ for every x , and if Σ_Y separates points then $\delta_y = \delta_{y'}$ forces $y = y'$.

(3) Let $(B_n)_n$ generate Σ_Y and let μ be $\{0, 1\}$ -valued. Put $C_n = B_n$ if $\mu(B_n) = 1$ and $C_n = Y \setminus B_n$ otherwise, and $C = \bigcap_n C_n$; then $\mu(C) = 1$, so $C \neq \emptyset$. If $y, y' \in C$, the class of those $B \in \Sigma_Y$ containing both or neither of y, y' is a σ -algebra containing every B_n , hence all of Σ_Y ; as Σ_Y separates points, $y = y'$. Thus $C = \{y\}$, so $\mu(\{y\}) = 1$ and $\mu = \delta_y$, with y unique because Σ_Y separates points. For the consequence: by (1) each $k(x)$ is $\{0, 1\}$ -valued, hence $\delta_{f(x)}$ for a unique $f(x)$; f is measurable because $f^{-1}(B) = \{x : k(x)(B) = 1\} \in \Sigma_X$ for every $B \in \Sigma_Y$; and $k = \delta f$ by definition of δ . Uniqueness of f is (2).

(4) This holds in any Markov category. From $v \circ u = \text{id}_X$ and determinism of v , $\text{copy}_X = \text{copy}_X \circ v \circ u = (v \circ v) \circ \text{copy}_A \circ u$, hence $(u \otimes u) \circ \text{copy}_X = ((u \circ v) \otimes (u \circ v)) \circ \text{copy}_A \circ u = \text{copy}_A \circ u$ by $u \circ v = \text{id}_A$. $\square$

Lemma A.2 (Ulam). *Call a countably additive $\{0, 1\}$ -valued probability measure on $(A, 2^A)$ that is not a Dirac measure a two-valued measure on the set A .*

1. *Two-valued measures on A correspond bijectively to countably complete non-principal ultrafilters on A [25, (10.3)–(10.4)].*
2. *If A carries no two-valued measure, then neither does any set of cardinality at most $|A|$, nor the power set 2^A [37, Bem. 2 and Satz 1]. In particular no set of cardinality at most that of the continuum carries one.*
3. *In modern terms, A carries a two-valued measure if and only if there is a measurable cardinal $\kappa \leq |A|$ [25, Lemma 10.2 and Def. 10.3]; Ulam's original form is that A carries none when no weakly inaccessible cardinal is $\leq |A|$ [37, Satz (A)], cf. [25, Thm. 10.1].*

Proof. (1) For a $\{0, 1\}$ -valued μ the class $\mathcal{U} = \{B \subseteq A : \mu(B) = 1\}$ is an ultrafilter on A , countably complete by countable additivity, and non-principal exactly when μ is not Dirac; conversely a countably complete non-principal ultrafilter $\mathcal{U}$ defines such a μ by $\mu(B) = 1$ iff $B \in \mathcal{U}$.

(2) If $i: B \rightarrow A$ is injective and μ is a two-valued measure on B , then $C \mapsto \mu(i^{-1}(C))$ is a two-valued measure on A : it is countably additive, gives A mass $\mu(B) = 1$, and vanishes on singletons because $i^{-1}(\{a\})$ has at most one element. This proves the first assertion. For the power set, identify 2^A with the set of functions $A \rightarrow \{0, 1\}$ and suppose ν is a two-valued measure on it. For each $a \in A$ exactly one of the two sets $\{g : g(a) = 0\}, \{g : g(a) = 1\}$ has ν -measure 1; call it E_a . We claim that $\nu(\bigcap_a E_a) = 1$. Otherwise $\nu(\bigcup_a (2^A \setminus E_a)) = 1$; well-order A as $(a_\xi)_{\xi < \lambda}$ and put $D_\xi := (2^A \setminus E_{a_\xi}) \setminus \bigcup_{\eta < \xi} (2^A \setminus E_{a_\eta})$, pairwise disjoint sets of ν -measure 0 with union $\bigcup_a (2^A \setminus E_a)$. Then $C \mapsto \nu(\bigcup_{\xi \in C} D_\xi)$ is a two-valued measure on λ , hence on A by the first assertion, contrary to hypothesis. But $\bigcap_a E_a$ is the single function g_0 with $g_0(a)$ the value singled out by E_a , so $\nu(\{g_0\}) = 1$, contradicting non-principality. Finally, a countable set carries no two-valued measure, since countable additivity and vanishing on singletons would force total mass 0; so neither does $2^{\aleph_0}$, nor any set of cardinality at most $2^{\aleph_0}$.

(3) If A carries a countably complete non-principal ultrafilter, so does the cardinal $|A|$, by transport along a bijection; the least cardinal κ carrying one is then $\leq |A|$, and it is measurable because such an ultrafilter on it is automatically κ -complete [25, Lemma 10.2]. Conversely a measurable $\kappa \leq |A|$ carries a κ -complete, hence countably complete, non-principal ultrafilter, which transports to A by the injection argument of (2). Ulam's form is subsumed by the modern one, since every measurable cardinal is strongly inaccessible [25, Lemma 10.4]. $\square$

Hypothesis (a) of Proposition 4.11 is thus the statement that no measurable cardinal is $\leq |\Theta'_0|$. By Lemma A.2(2) it holds outright, with no large-cardinal input, whenever Θ'_0 has at most the cardinality of

the continuum, which covers every parameter set in this paper, and it holds for every set if measurable cardinals do not exist.

Proof of Proposition 4.11. The functor. On objects, $\Delta(\Theta_0, S, p) = \widehat{p}$. On a morphism $(r, f): (\Theta_0, S, p) \rightarrow (\Theta'_0, S', p')$, a function $r: \Theta_0 \rightarrow \Theta'_0$ and a measurable $f: S \rightarrow S'$ with $\mathcal{P}(f) \circ p = p' \circ r$ (Definition 2.6), put $\Delta(r, f) = (\delta r, \delta f)$. Here r is measurable because its domain is discrete, so both components are Dirac kernels, hence deterministic (Definition 3.2); by Lemma 4.9 the pair is a square from $\widehat{p}$ to $\widehat{p'}$ iff $\widehat{p'}(r(\theta)) = f_* \widehat{p}(\theta)$ for every θ , i.e. iff $p' \circ r = \mathcal{P}(f) \circ p$, which is the hypothesis. Since δ is a functor and composition in $\text{Stoch}_{\text{det}}^{\rightarrow}$ is componentwise, Δ preserves identities and composites.

Part (1). By the conventions above, $(\Theta_0, S, p) \mapsto \widehat{p}$ is a bijection from the objects of StatMod onto the arrows of Stoch whose domain carries its full power set, with inverse $q \mapsto (A, S, \check{q})$ for $q: A \cdot I \rightsquigarrow S$: the domain of $\widehat{p}$ determines Θ_0 as its underlying set, the codomain determines S , and $p = \check{p}$. In particular Δ is injective on objects and lands in $\text{Stoch}_{\text{det, cop}}^{\rightarrow}$. For essential surjectivity onto $\text{Stoch}_{\text{det, cop}}^{\rightarrow}$, let $q: X \rightsquigarrow S$ with $X \cong A \cdot I$ in Stoch , say via mutually inverse $v: A \cdot I \rightsquigarrow X$ and $u: X \rightsquigarrow A \cdot I$. We show that both are deterministic, arguing from the copower side, where singletons are measurable. For $a \in A$ put $X_a := \{x \in X : u(x)(\{a\}) = 1\}$, measurable since $x \mapsto u(x)(\{a\})$ is. Since $u \circ v = \text{id}_{A \cdot I}$,

$$1 = \delta_a(\{a\}) = (u \circ v)(a)(\{a\}) = \int_X u(x)(\{a\}) v(a)(dx),$$

and the integrand is bounded by 1, so $v(a)(X_a) = 1$; in particular $X_a \neq \emptyset$. For $x \in X_a$ the probability measure $u(x)$ on 2^A gives mass 1 to $\{a\}$, so $u(x) = \delta_a$, and then, since $v \circ u = \text{id}_X$,

$$\delta_x(F) = (v \circ u)(x)(F) = \int_A v(b)(F) u(x)(db) = v(a)(F) \quad (F \in \Sigma_X).$$

Thus $v(a) = \delta_x$ for every $x \in X_a$. Choosing $x_a \in X_a$ for each a (axiom of choice) gives a function $f: A \rightarrow X$, measurable because its domain is discrete, with $v = \delta f$; so v is deterministic, and u is deterministic by Lemma A.1(4). Now $q \circ v: A \cdot I \rightsquigarrow S$ has domain $A \cdot I$, so $q \circ v = \Delta(A, S, (q \circ v)^\vee)$; and $(v, \text{id}_S): q \circ v \rightarrow q$ and $(u, \text{id}_S): q \rightarrow q \circ v$ are squares, $q \circ v = \text{id}_S \circ (q \circ v)$ and $(q \circ v) \circ u = q = \text{id}_S \circ q$, which are mutually inverse in $\text{Stoch}_{\text{det}}^{\rightarrow}$, since $(u, \text{id}_S) \circ (v, \text{id}_S) = (u \circ v, \text{id}_S) = (\text{id}, \text{id})$ and symmetrically. Hence $q \cong \Delta(A, S, (q \circ v)^\vee)$.

Part (2). Fix $x = (\Theta_0, S, p)$ and $x' = (\Theta'_0, S', p')$ with x' satisfying (a) and (b), and let (a, b) be a square from $\widehat{p}$ to $\widehat{p'}$. Its left leg $a: \Theta_0 \cdot I \rightsquigarrow \Theta'_0 \cdot I$ is deterministic, so by Lemma A.1(1) each $a(\theta)$ is a countably additive $\{0, 1\}$ -valued probability measure on $2^{\Theta'_0}$; by (a) it is $\delta_{r(\theta)}$ for a unique $r(\theta) \in \Theta'_0$, and r is measurable because Θ_0 is discrete, so $a = \delta r$. The right leg is $b = \delta f$ for a unique measurable $f: S \rightarrow S'$, by (b) for S' and Lemma A.1(3). By Lemma 4.9, $(\delta r, \delta f)$ is a square iff $p' \circ r = \mathcal{P}(f) \circ p$, so (r, f) is a morphism of StatMod with $\Delta(r, f) = (a, b)$: the map $\text{StatMod}(x, x') \rightarrow \text{Stoch}_{\text{det}}^{\rightarrow}(\Delta x, \Delta x')$ is surjective. It is injective by Lemma A.1(2), since $2^{\Theta'_0}$ separates points and $\Sigma_{S'}$ does by (b).

Part (3). Write $\mathcal{A}$ for the full subcategory of $\text{Stoch}_{\text{det, cop}}^{\rightarrow}$ described in the statement. For $x = (\Theta_0, S, p)$ in $\text{StatMod}_{(a), (b)}$ the object $\Delta x = \widehat{p}$ has domain $\Theta_0 \cdot I$ with Θ_0 satisfying (a) and codomain S satisfying (b), so Δ restricts to a functor $\text{StatMod}_{(a), (b)} \rightarrow \mathcal{A}$. It is fully faithful by part (2), because every object of $\text{StatMod}_{(a), (b)}$ satisfies (a) and (b), and it is essentially surjective by the argument of part (1): for $q: X \rightsquigarrow S$ in $\mathcal{A}$, with $v: A \cdot I \rightsquigarrow X$ an isomorphism and A satisfying (a) as in the definition of $\mathcal{A}$, the object $(A, S, (q \circ v)^\vee)$ lies in $\text{StatMod}_{(a), (b)}$ and $q \cong \Delta(A, S, (q \circ v)^\vee)$. A fully faithful and essentially surjective functor is an equivalence [27, IV.4, Thm. 1(iii)].

The finite case. Every object of FinStoch is $A \cdot I$ for a finite set A , so the object bijection of part (1) restricts to a bijection from the objects of $\text{StatMod}_{\text{fin}}$ onto the morphisms of FinStoch , i.e. onto the objects of $\text{FinStoch}_{\text{det}}^{\rightarrow}$ (which, FinStoch being full in Stoch , is the full subcategory of $\text{Stoch}_{\text{det}}^{\rightarrow}$ on these objects). Conditions (a) and (b) hold for every finite set and every finite discrete space, so by part (2) Δ is bijective on every hom-set. A functor bijective on objects and on hom-sets is an isomorphism of categories, so $\text{StatMod}_{\text{fin}} \cong \text{FinStoch}_{\text{det}}^{\rightarrow}$, which is more than the stated equivalence. $\square$

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